

July 7, 2026 05:58 PM GMT

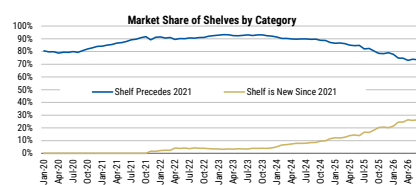
ABS Strategy | North America

ABS Dashboard July 2026: Spotlight on Unsecured Consumer Vintage Performance

Key Takeaways

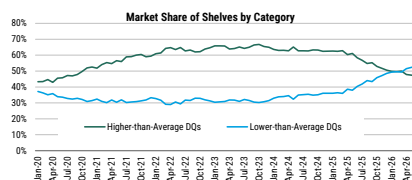
- Over 2Q26, \$117bn of ABS was issued, a new record. Consumer ABS spreads finally softened in June, even as supply eased marginally m/m to \$37bn.
- Prime auto DQs held steady in June at 0.48%. In subprime, DQs ticked up to 5.47% in line with historical seasonal behavior.
- In Consumer MPL ABS, the 2025 vintage continues to be the strongest performing and aggregate delinquencies continue to soften.
- We dig into this asset class in our Spotlight, finding that this strength in aggregate performance has less to do with individual performance improvements.
- Rather, we find that the introduction of new, lower-than-average delinquency shelves since 2021 is meaningfully contributing to the aggregate's improvements.

Exhibit 1: Shelves that are new since 2021 now account for roughly 26% of the outstanding balance...



Source: Intex, Morgan Stanley Research. Our universe is of the 50 largest shelves today.

Exhibit 2: ... and this pickup in market-share coincides with a pickup in market-share of lower-than-average DQ shelves.



Source: Intex, Morgan Stanley Research. Our universe is of the 50 largest shelves today.

Please [email us](#) for a copy of the dashboard data in this report.

MORGAN STANLEY & CO. LLC

Carolyn L. Campbell

Strategist

Carolyn.Campbell@morganstanley.com

+1 212 761-3748

James Egan

Strategist

James.F.Egan@morganstanley.com

+1 212 761-4715

Raquel Kanner

Strategist

Raquel.Kanner@morganstanley.com

+1 212 761-1103

Morgan Stanley | RESEARCH

We appreciate your support

**2026 EXTEL GLOBAL
FIXED INCOME POLL**

VOTE NOW

★★★★★

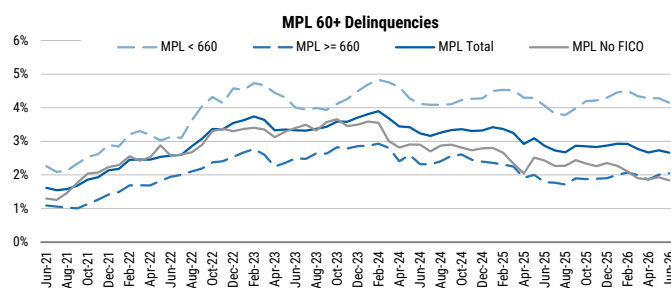
Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

Spotlight on Unsecured Consumer Vintage Performance

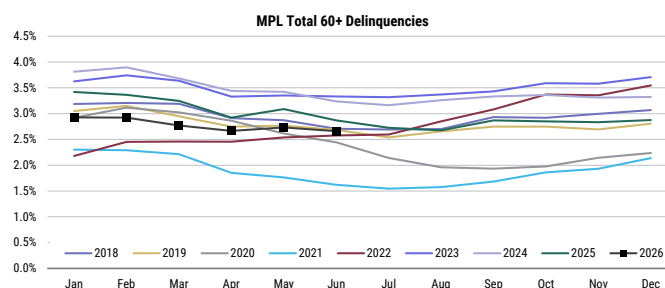
The performance data of unsecured consumer/MPL ABS looks strong this year, a different story than in auto ABS which continues to show weakness vs prior years. Aggregate delinquencies in MPL have been falling for a number of years, with 60+ DQs now well below prior years' highs. Easing of lending standards during Covid saw performance deteriorate rapidly. A common heuristic has been that lenders rapidly tightened standards in the aftermath, and to be sure that is indeed what reporting suggests (see [Exhibit 7](#)). However, we find that improved 2025 vintage performance is a more nuanced story, with changes in the universe as the MPL ABS market has grown contributing meaningfully as well.

Exhibit 3: 60+ delinquencies for unsecured consumer/MPL ABS



Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 4: MPL 60+ DQs, compared annually

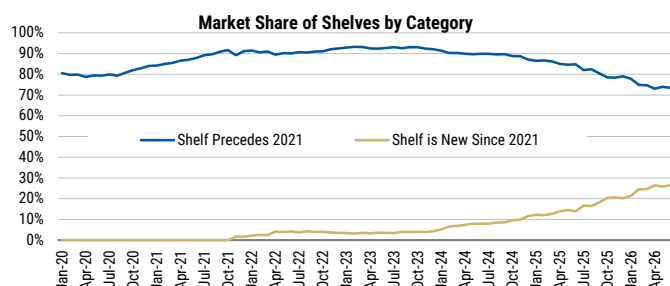


Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Like with subprime autos, the improved MPL performance is not just due to individual lender behavior changes, but also due to a *change* in the universe composition. Unlike autos, however, this universe mix works in the favor of the aggregate. Of the largest 25 shelves today, ten are new since 2021, accounting for 22% of the total universe, and all of those ten have lower-than-average delinquencies.

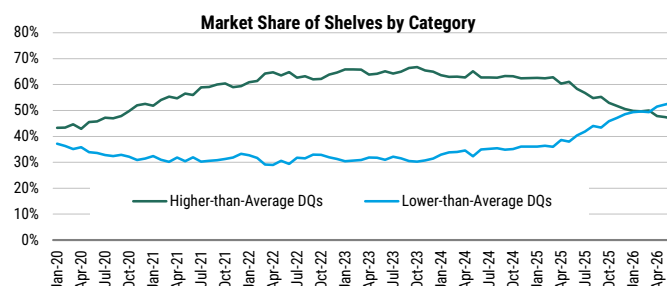
Overall, shelves that launched since the beginning of 2021 account now for just over a quarter of the outstanding balance of unsecured consumer, with a real acceleration of market share acquisition since the start of 2024. (We base our analysis on the top 50 MPL shelves by current balance). At the same time, the share of shelves with lower-than-average 60+ delinquencies has been steadily increasing. We define lower-than-average shelves as those for whom their average DQ rate over the past 12 months is below the average DQ rate of the universe over the same time period, and the higher-than-average shelves as the converse.

Exhibit 5: Shelves that are new since 2021 now account for roughly 26% of the outstanding balance...



Source: Intex, Morgan Stanley Research

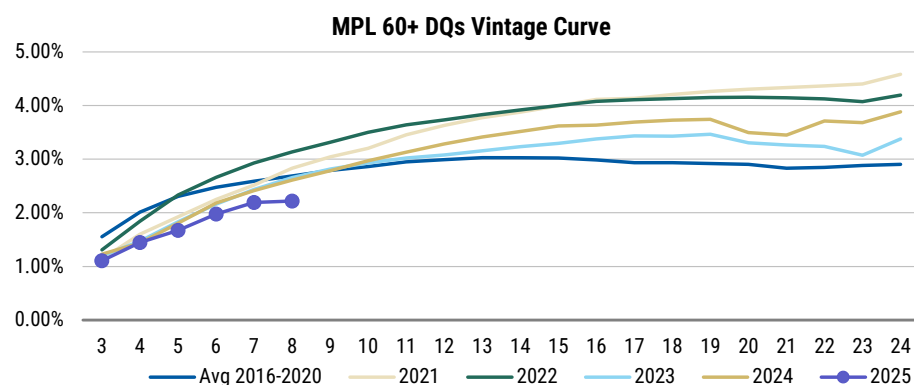
Exhibit 6: ... and this pickup in market-share coincides with a pickup in market-share of lower-than-average DQ shelves.



Source: Intex, Morgan Stanley Research

We think this change in issuers helps explain why the 2025 vintage curve for MPL looks so strong compared to prior years, but when we investigate shelves individually we do not find a consistent trend of improvement. Indeed, the 2025 vintage is the strongest in the MPL universe vs every prior year, including the average of 2016-2020 (Exhibit 7). However, when we look at individual shelves, the pattern is less clear. In fact, none of the largest shelves (defined now as those with > \$1bn currently outstanding) have, at 8 months of seasoning, better delinquency performance than all other years as we see with the aggregate (Exhibit 8). (Please email us if you would like to see individual vintage curves for the largest issuers. We include below a sample of the largest six shelves).

Exhibit 7: The MPL 2025 Vintage is better performing than all prior years



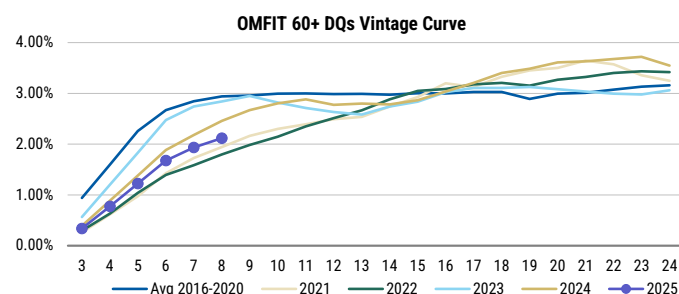
Source: Intex, Morgan Stanley Research

Exhibit 8: The performance at the individual shelf level is more mixed. Some do demonstrate stronger performance vs prior years, but not as consistently as the aggregate suggests.

8-Month 60+ DQs Vintage Comparison						
	2025 vs 2024	2025 vs 2023	2025 vs 2022	2025 vs 2021	2025 vs Avg 2016-2020	
OMFIT	-0.34%	-0.72%	0.32%	0.17%	-0.82%	
PAID	0.13%	0.25%	0.08%	-0.19%	1.39%	
AFMRT		-0.03%	0.28%	0.45%	0.46%	
SLCLC	0.06%	0.09%				
LFT	0.49%	-0.43%	-0.43%	1.75%	1.62%	
MFIT	0.36%	0.40%	0.89%	0.79%	1.16%	
OPTN	-0.54%		-0.33%	0.11%	0.42%	
AFFRM	1.03%	1.31%	0.94%	1.56%	2.33%	
UPST	0.30%	-0.53%	-1.97%	0.30%	1.28%	
BHG	0.11%	-0.38%	-0.99%	-0.50%	0.14%	
UMPT		0.30%		0.78%	0.60%	
REACH	-0.41%	-0.94%	-0.44%	0.67%	0.77%	
RMIT	-0.02%		0.32%	0.31%	1.45%	
CHRY	-0.01%					
REPS	1.54%			1.62%	4.02%	
LENDR						
Overall	-0.39%	-0.44%	-0.92%	-0.61%	-0.47%	

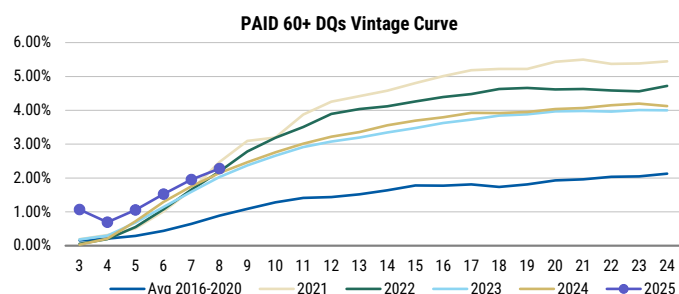
Source: Intex, Morgan Stanley Research

Exhibit 9: OMFIT Vintage Curve



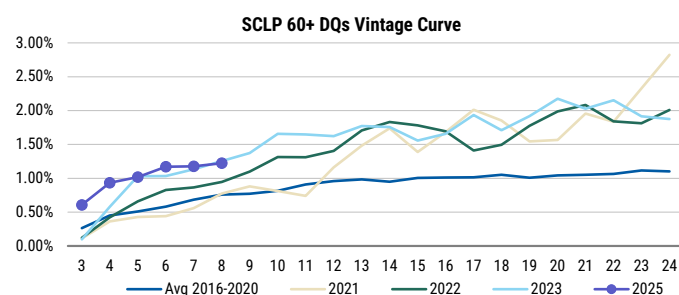
Source: Intex, Morgan Stanley Research

Exhibit 10: PAID Vintage Curves



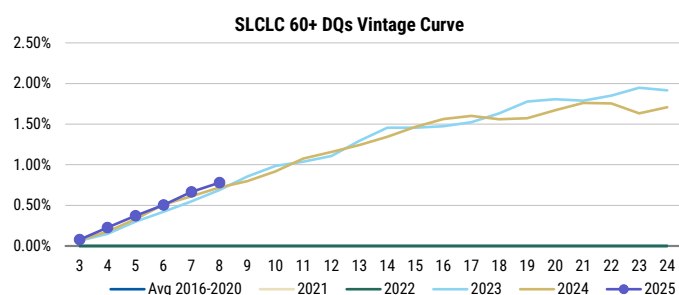
Source: Intex, Morgan Stanley Research

Exhibit 11: SCLP Vintage Curves

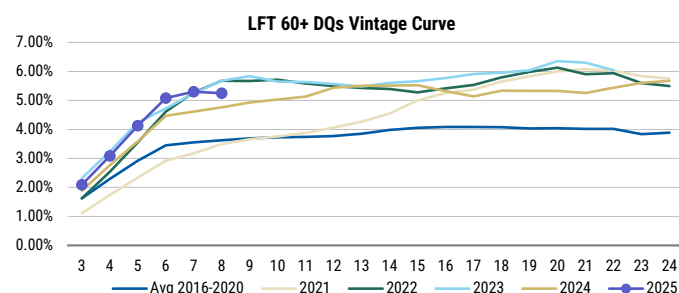
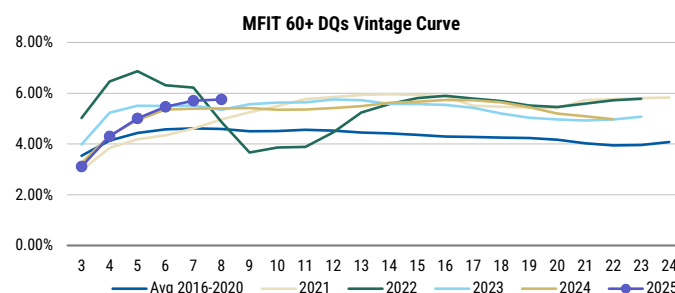


Source: Intex, Morgan Stanley Research

Exhibit 12: SLCLC Vintage Curves



Source: Intex, Morgan Stanley Research

Exhibit 13: LFT Vintage Curves**Exhibit 14:** MFIT Vintage Curves

In essence, individual vintage curves fail to demonstrate a consistent pattern of tighter lending standards. Rather, we find that the introduction of *new* shelves with lower-than-average delinquency trends is playing a key role in bringing the aggregate curve lower.

Exhibit 15: Of the largest MPL shelves today, ten have been introduced since 2021 and all of those shelves have lower-than-average delinquencies.

Top 25 MPL Shelves by Current Outstanding Balance						
Shelf	New Since 2021?	LTM Average 60+ DQ	LTM 60+ DQs vs MPL Average	Current Market Share	Change in Market Share since Jan 2021	Current Outstanding Balance (\$mm)
LENDR	Yes	0.39%	0.1x	1.29%	1.29%	\$727
BOLT	Yes	0.40%	0.1x	0.75%	0.75%	\$419
AQFIT	Yes	0.71%	0.3x	0.71%	0.71%	\$397
SCPL	No	1.01%	0.4x	6.47%	-4.09%	\$3,636
SLCLC	Yes	1.13%	0.4x	6.27%	6.27%	\$3,528
RKTL	Yes	1.19%	0.4x	0.63%	0.63%	\$352
AFRMT	Yes	1.30%	0.5x	8.48%	8.48%	\$4,769
CHRY	Yes	1.35%	0.5x	1.55%	1.55%	\$873
UMPT	No	1.53%	0.5x	2.99%	2.13%	\$1,682
REACH	No	1.57%	0.6x	2.90%	2.19%	\$1,628
BHG	No	1.57%	0.6x	3.05%	2.48%	\$1,715
BEAST	Yes	1.62%	0.6x	1.07%	1.07%	\$604
POSH	Yes	1.64%	0.6x	0.71%	0.71%	\$401
RCKTL	Yes	1.78%	0.6x	0.68%	0.68%	\$382
AFFRM	No	1.81%	0.6x	2.71%	-0.39%	\$1,524
AVNT	No	2.37%	0.8x	1.59%	-0.04%	\$894
OPTN	No	2.39%	0.9x	4.08%	-0.86%	\$2,297
UPST	No	2.46%	0.9x	3.53%	-1.47%	\$1,986
MFT	No	2.78%	1.0x	1.04%	-3.58%	\$583
PAID	No	2.93%	1.0x	12.78%	9.99%	\$7,187
OMFIT	No	3.64%	1.3x	14.93%	-5.56%	\$8,397
RMIT	No	4.25%	1.5x	2.40%	0.80%	\$1,351
LFT	No	5.14%	1.8x	5.27%	-0.78%	\$2,964
MFIT	No	5.20%	1.9x	5.19%	2.99%	\$2,921
REPS	No	8.17%	2.9x	2.15%	0.37%	\$1,212

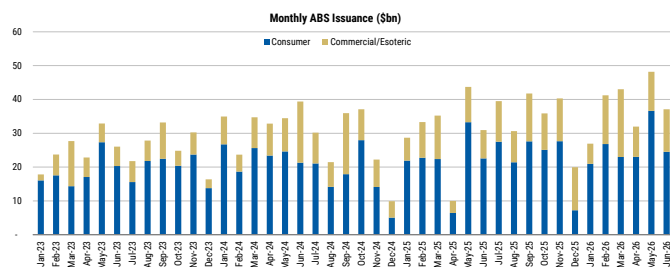
Source: Intex, Morgan Stanley Research

New Issuance, Outstanding Balance, and Trading Volumes

ABS issuance volumes set a new record in 2Q26 with \$117bn issued, +6% over 1Q26, bringing the YTD totals to \$228bn (+26% vs 2025 YTD). June was a slightly slower month than May, with \$37bn issued, but still 20% higher than June 2025 volumes and with a decent mix of both consumer (\$24.5bn) and commercial supply (\$12.5bn). A handful of digital infrastructure deals also priced in June for July settle, ending a 2-month drought in data centers with new deals from QTS and Aligned and bringing fiber YTD totals to nearly \$10bn with deals from Uniti and FirstLight. New issue spreads in both data centers and fiber tightened vs previous deals across the stack, with demand enthusiastic.

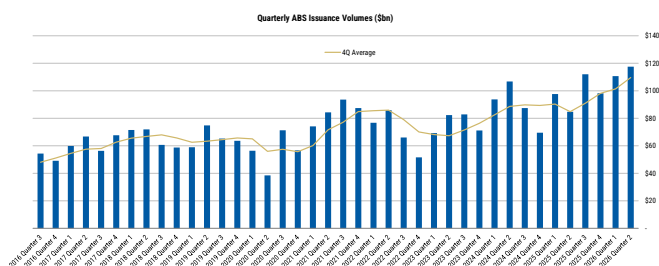
Consumer ABS spreads finally halted their rally with prime auto AAA widening ~8bps vs the May tightens and subprime auto AAA ~13bps wider. We were surprised last month that spreads continued to hit new YTD tightens despite the very heavy supply; the ongoing heavy issuance, not just in consumer ABS but in resi, helped to finally push spreads slightly wider across the consumer sec credit complex.

Exhibit 16: Consumer and commercial ABS volumes by month



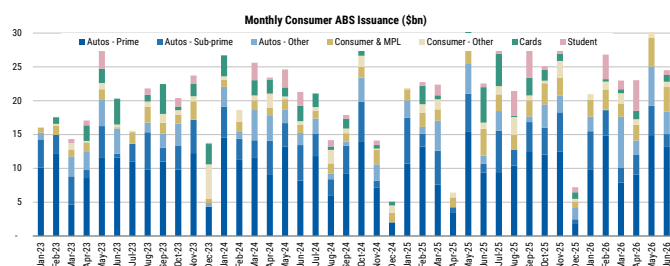
Source: Bloomberg, Morgan Stanley Research

Exhibit 17: Quarterly US ABS issuance volumes and rolling four-quarter average



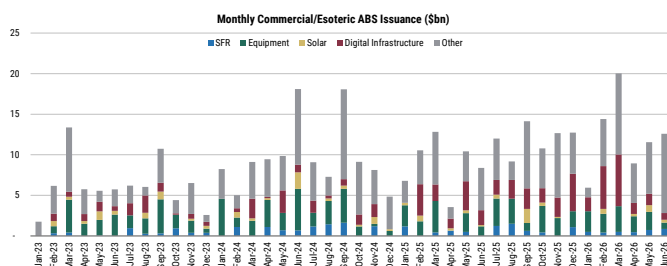
Source: Bloomberg, Morgan Stanley Research

Exhibit 18: Consumer ABS issuance volumes by month and subcategory



Source: Bloomberg, Finsight, Morgan Stanley Research

Exhibit 19: Commercial/esoteric ABS issuance volumes by month and subcategory



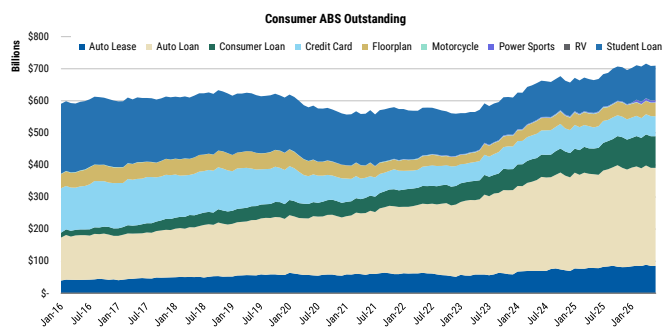
Source: Bloomberg, Finsight, Morgan Stanley Research

Exhibit 20:

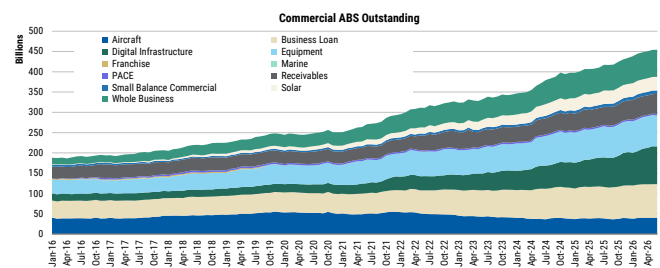
ABS issuance volumes versus historicals

Type	Monthly Issuance					YTD Issuance									
	Jun-26	May-26	m/m	Jun-25	y/y	2026	2025	2024	2023	2022	2021	2020	2019		
Autos - Prime	\$ 13.2	\$ 14.9	-12%	\$ 9.4	40%	\$ 69.6	\$ 59.7	\$ 68.0	\$ 58.8	\$ 39.9	\$ 43.8	\$ 34.9	\$ 28.7		
Autos - Sub-prime	\$ 1.9	\$ 4.3	-56%	\$ 1.3	48%	\$ 20.9	\$ 21.4	\$ 23.9	\$ 17.4	\$ 21.9	\$ 20.3	\$ 12.7	\$ 16.8		
Consumer & MPL	\$ 4.4	\$ 6.3	-30%	\$ 4.9	-9%	\$ 24.9	\$ 19.4	\$ 14.0	\$ 9.1	\$ 14.1	\$ 11.7	\$ 6.7	\$ 11.7		
Cards	\$ 1.0	\$ 4.2	-76%	\$ 5.3	-81%	\$ 7.4	\$ 11.5	\$ 11.1	\$ 9.2	\$ 15.1	\$ 2.8	\$ 1.3	\$ 8.8		
Student	\$ 0.7	\$ 1.0	-30%	\$ 0.5	32%	\$ 11.1	\$ 3.5	\$ 7.5	\$ 3.9	\$ 5.3	\$ 18.6	\$ 10.0	\$ 8.9		
Equipment	\$ 0.8	\$ 2.2	-65%	\$ 1.2	-34%	\$ 12.1	\$ 11.8	\$ 17.7	\$ 10.3	\$ 10.9	\$ 10.6	\$ 9.1	\$ 10.6		
SFR	\$ 0.9	\$ 0.7	15%	\$ -	-	\$ 3.5	\$ 2.8	\$ 3.6	\$ 0.8	\$ 9.8	\$ 3.8	\$ 2.9	\$ 1.2		
Digital Infrastructure	\$ 0.8	\$ 1.4	-41%	\$ 1.8	-55%	\$ 16.9	\$ 12.5	\$ 6.7	\$ 4.1	\$ 1.7	\$ 1.7	\$ 0.8	\$ 1.0		
Solar	\$ 0.4	\$ 0.8	-57%	\$ 0.2	139%	\$ 2.1	\$ 1.8	\$ 3.2	\$ 2.8	\$ 2.9	\$ 2.4	\$ 1.2	\$ 0.9		
Other	\$ 13.1	\$ 12.2	8%	\$ 6.4	103%	\$ 59.9	\$ 37.3	\$ 44.3	\$ 34.4	\$ 41.3	\$ 42.8	\$ 15.5	\$ 45.0		
Total	\$ 37.1	\$ 48.2	-23%	\$ 30.9	20%	\$ 228.4	\$ 181.8	\$ 200.1	\$ 151.0	\$ 162.7	\$ 158.5	\$ 95.0	\$ 133.8		

Source: Bloomberg, Finsight, Morgan Stanley Research

Exhibit 21: Consumer ABS outstanding issuance

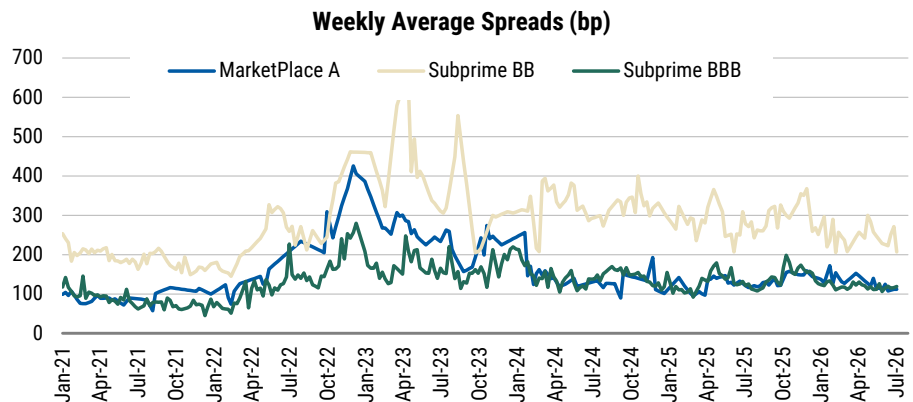
Source: Intex, Morgan Stanley Research

Exhibit 22: Commercial/esoteric ABS outstanding issuance

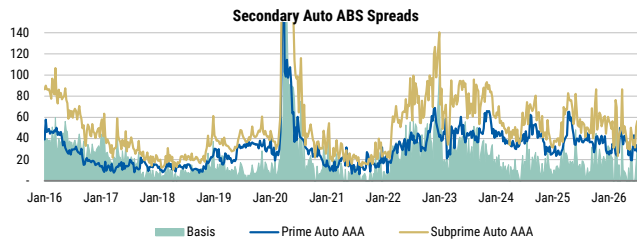
Source: Intex, Morgan Stanley Research

Exhibit 23:

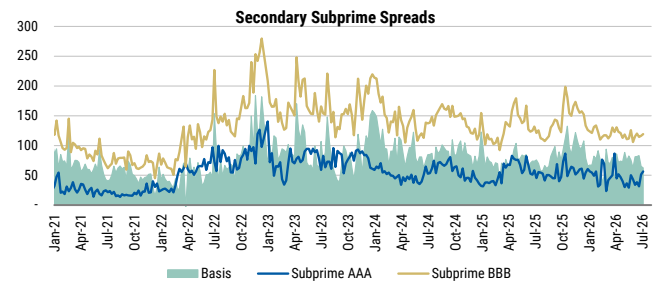
Weekly average spreads (bp) for marketplace and subprime auto



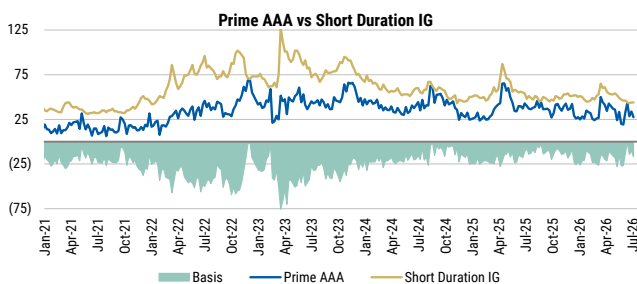
Source: TRACE, Morgan Stanley Research

Exhibit 24: Secondary Auto ABS Spreads

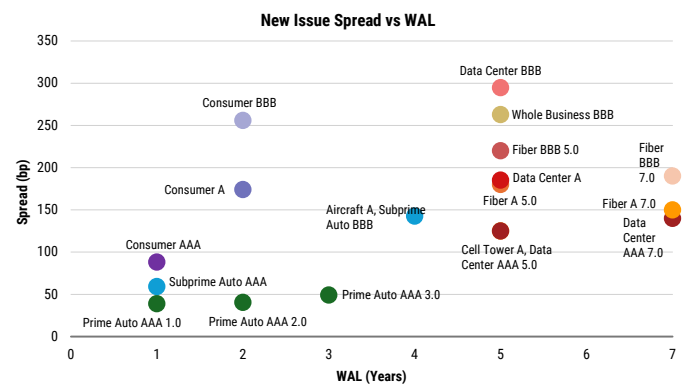
Source: TRACE, Morgan Stanley Research

Exhibit 25: Secondary Subprime Spreads

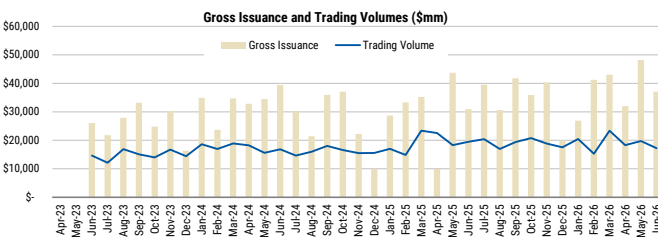
Source: TRACE, Morgan Stanley Research

Exhibit 26: Prime AAA vs Short Duration IG

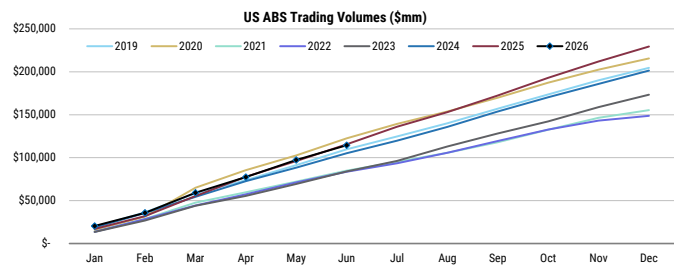
Source: TRACE, Bloomberg, Morgan Stanley Research

Exhibit 27: New Issue Spread vs WAL

Source: Finsight, Morgan Stanley Research

Exhibit 28: Trading volumes and gross issuance (\$mm)

Source: Bloomberg, TRACE, Morgan Stanley Research

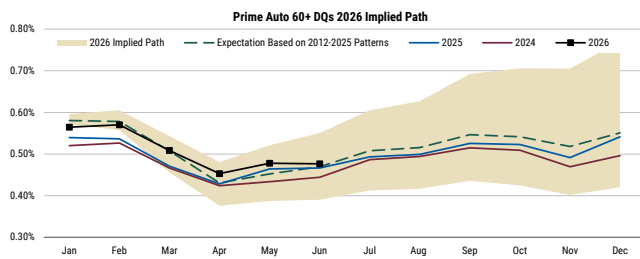
Exhibit 29: Cumulative trading volumes (\$mm)

Source: TRACE, Morgan Stanley Research

Performance of Auto Loans (Prime vs. Subprime)

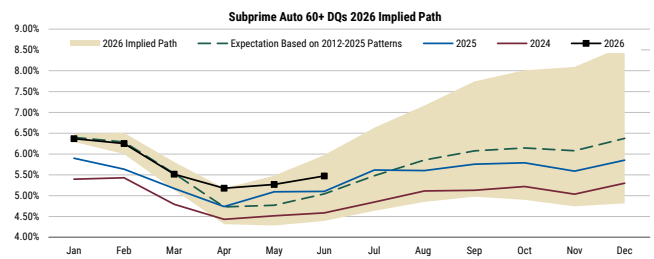
Prime delinquencies held steady in June at 0.48%, while subprime delinquencies rose 20bp to 5.47%, in line with typical seasonal trends for June. Cumulative default rates (CDR), on the other hand, saw improvements, with prime falling 1bp to 1.46% and subprime down 7bp to 12.67%. Losses remained stable in June, falling 1bp in both prime and subprime to 0.58% and 7.97%, respectively.

Exhibit 30: 60+ DQs in 2026 vs implied path, prime auto



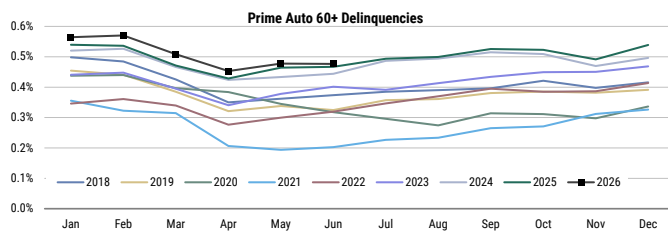
Source: Intex, Morgan Stanley Research

Exhibit 31: 60+ DQs in 2026 vs implied path, subprime auto



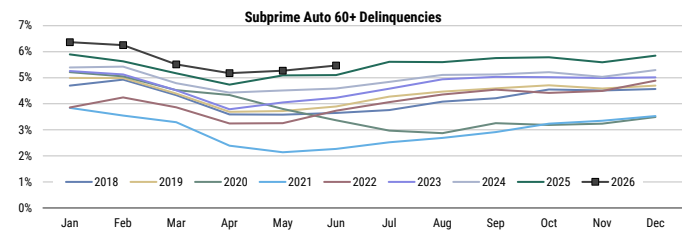
Source: Intex, Morgan Stanley Research

Exhibit 32: Prime auto 60+ delinquencies, annual comparison



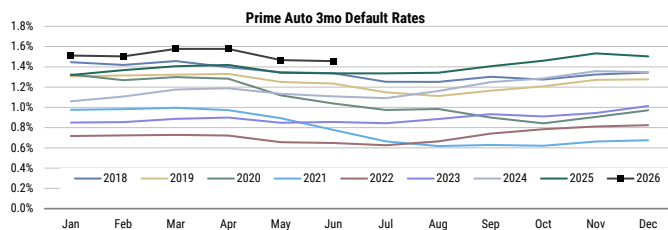
Source: Intex, Morgan Stanley Research

Exhibit 33: Subprime auto 60+ delinquencies, annual comparison



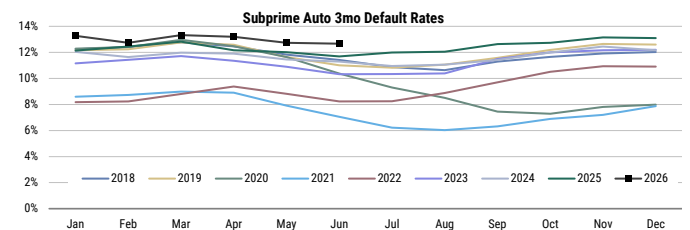
Source: Intex, Morgan Stanley Research

Exhibit 34: Prime auto 3mo default rates

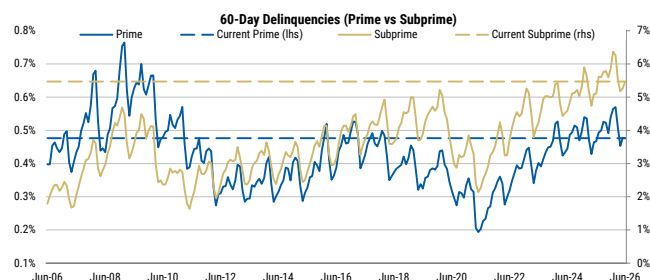


Source: Intex, Morgan Stanley Research

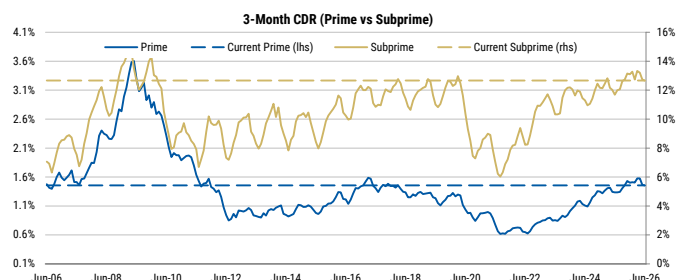
Exhibit 35: Subprime auto 3mo default rates



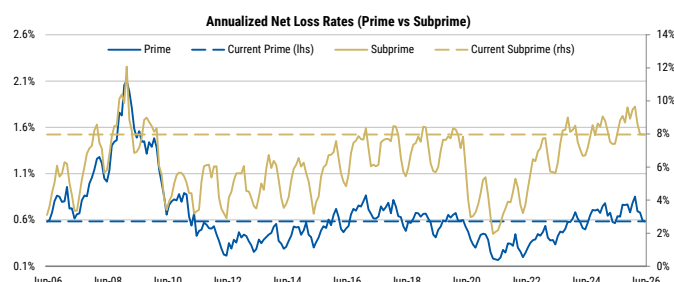
Source: Intex, Morgan Stanley Research

Exhibit 36: 60+ day delinquencies, prime versus subprime

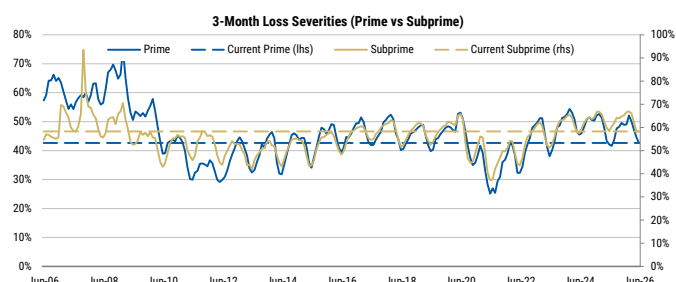
Source: Intex, Morgan Stanley Research

Exhibit 37: 3-month CDR, prime versus subprime

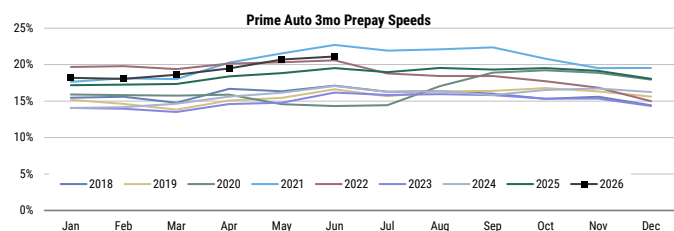
Source: Intex, Morgan Stanley Research

Exhibit 38: Annualized net loss rates, prime versus subprime

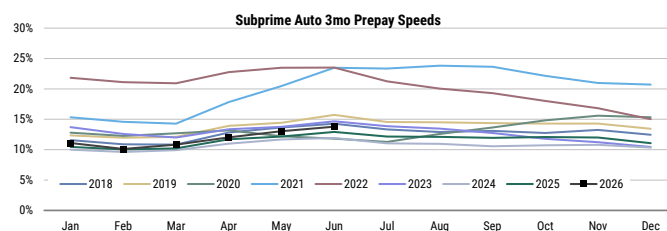
Source: Intex, Morgan Stanley Research

Exhibit 39: 3m loss severities, prime versus subprime

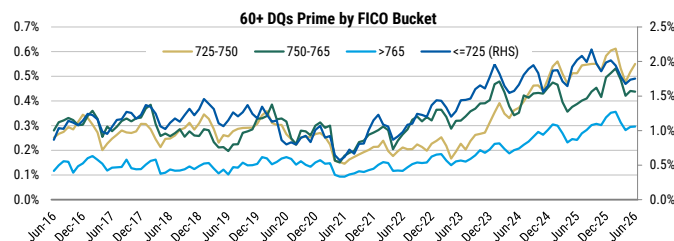
Source: Intex, Morgan Stanley Research

Exhibit 40: Prime auto 3mo prepay speeds

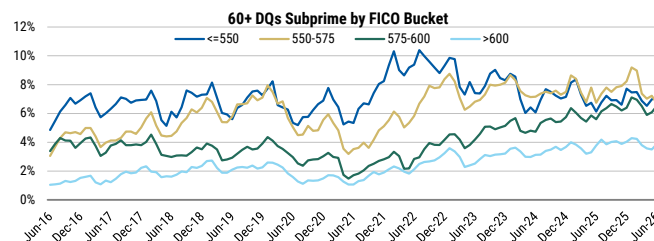
Source: Intex, Morgan Stanley Research

Exhibit 41: Subprime auto 3mo prepay speeds

Source: Intex, Morgan Stanley Research

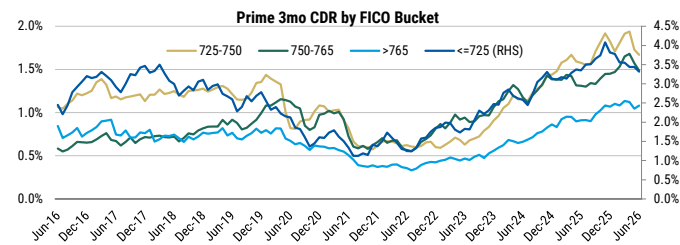
Exhibit 42: 60+ DQs prime by FICO bucket

Source: Intex, Company Deal Docs, Morgan Stanley Research

Exhibit 43: 60+ DQs subprime by FICO bucket

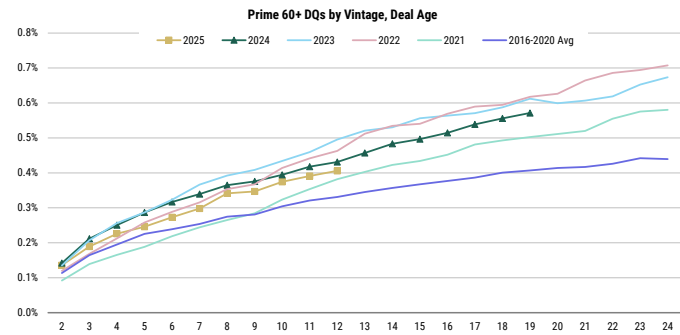
Source: Intex, Company Deal Docs, Morgan Stanley Research

Exhibit 44: Prime 3mo CDR by FICO bucket



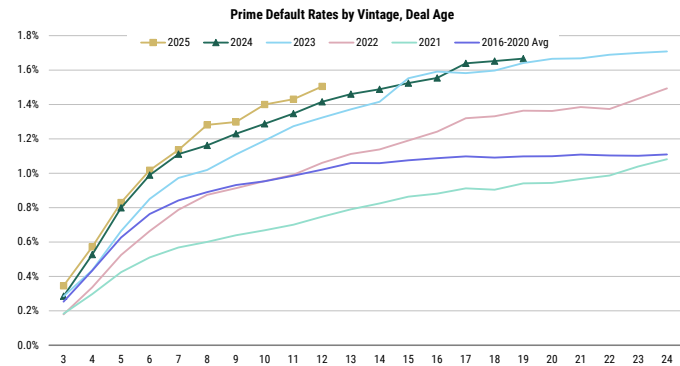
Source: Intex, Company Deal Docs, Morgan Stanley Research

Exhibit 46: Prime 60+ DQs by Vintage, Deal Age



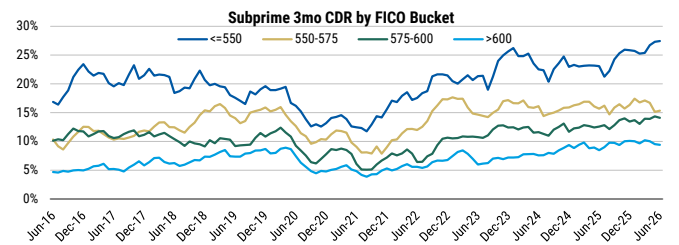
Source: Intex, Morgan Stanley Research

Exhibit 48: Prime Default Rates by Vintage, Deal Age



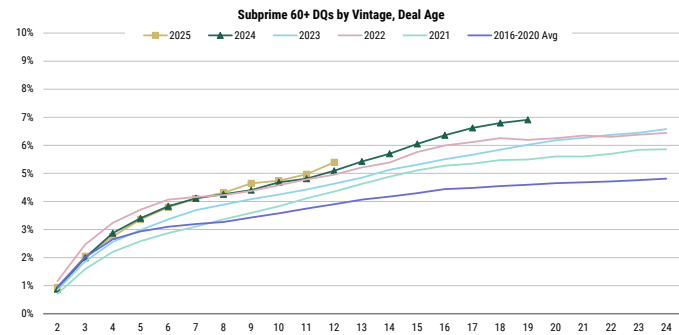
Source: Intex, Morgan Stanley Research

Exhibit 45: Subprime 3mo CDR by FICO bucket



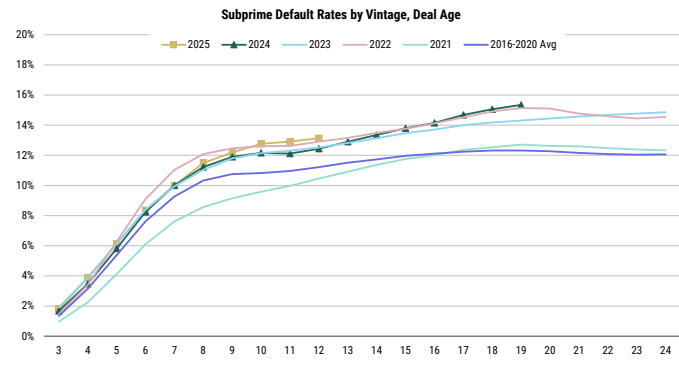
Source: Intex, Company Deal Docs, Morgan Stanley Research

Exhibit 47: Subprime 60+ DQs by Vint, Deal Age



Source: Intex, Morgan Stanley Research

Exhibit 49: Subprime Default Rates by Vintage, Deal Age

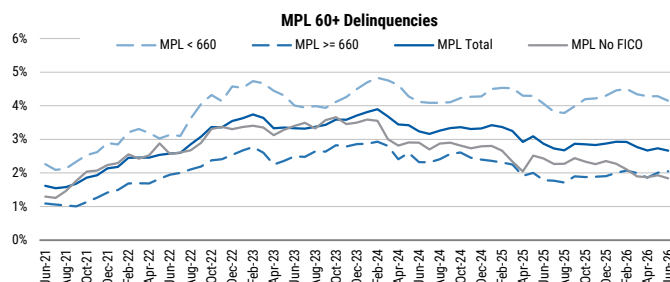


Source: Intex, Morgan Stanley Research

Performance of Marketplace Lending Deals

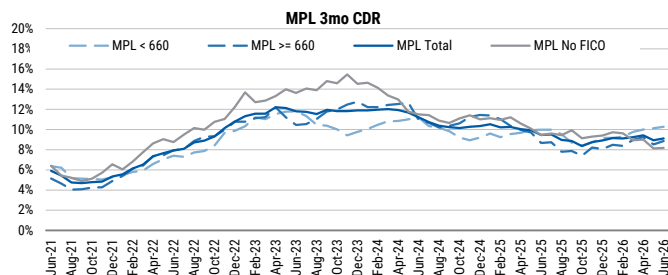
In June, MPL 60+ DQs saw a 7bp decrease to 2.66%, with the decline driven across high-FICO shelves which were down 13bp on the month to 4.15%. Low-FICO shelves, in contrast, rose 4bp to 2.05%. 3mo CDR slightly increased 17bp on the month from 8.94% to 9.11%. Prepayment speeds were up in June to 34.81%, from 34.50% in May.

Exhibit 50: 60+ delinquencies



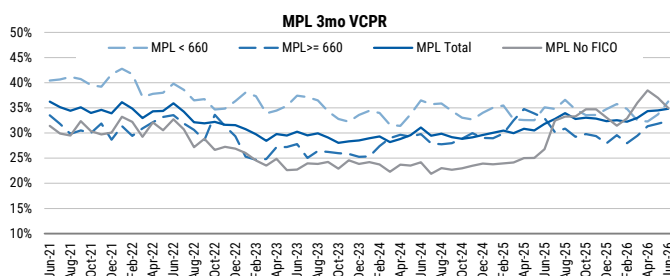
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 51: 3-month CDR



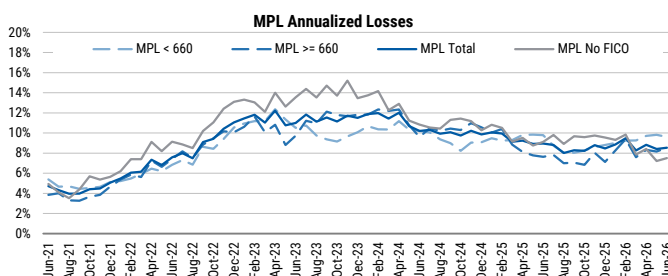
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 52: 3-month VPCR



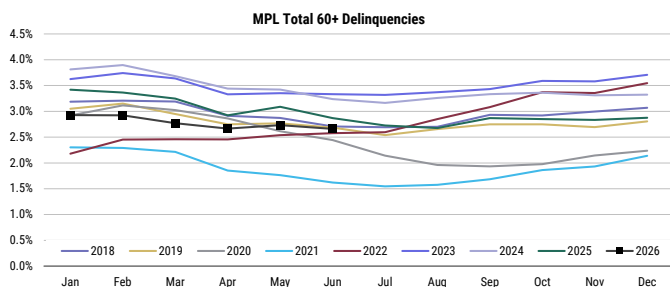
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 53: Annualized losses



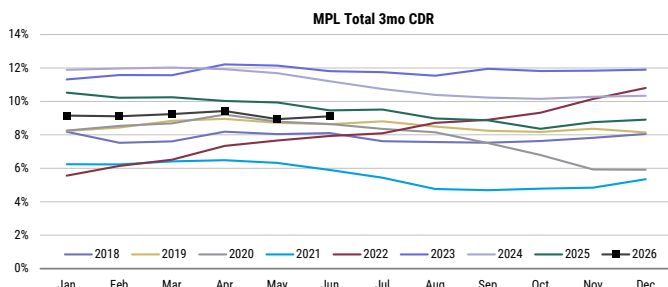
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 54: MPL Total 60+ DQs, annual comparison

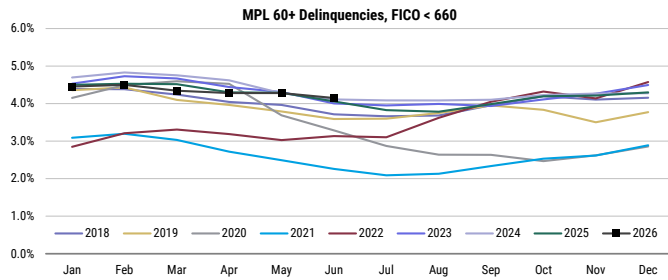


Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

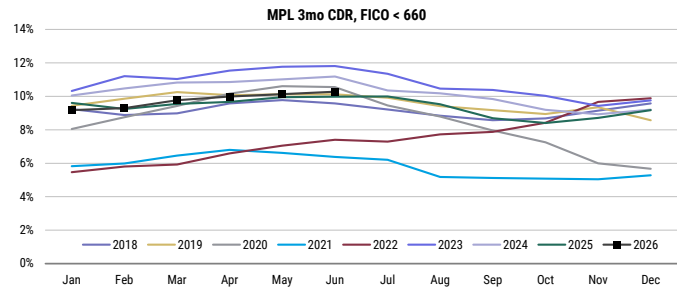
Exhibit 55: MPL Total defaults, annual comparison



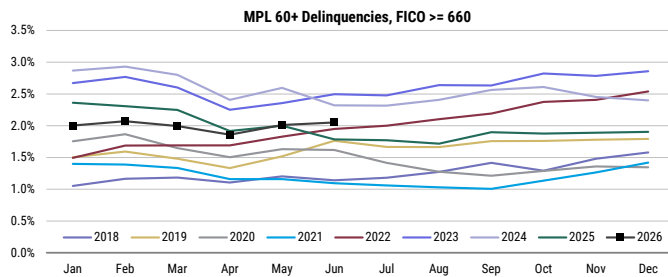
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 56: MPL FICO < 660 60+ DQs, annual comparison

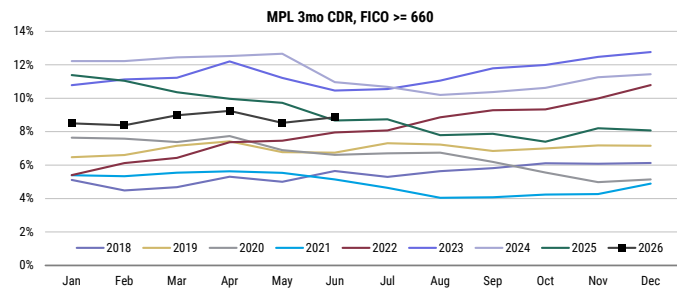
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 57: MPL FICO < 660 defaults, annual comparison

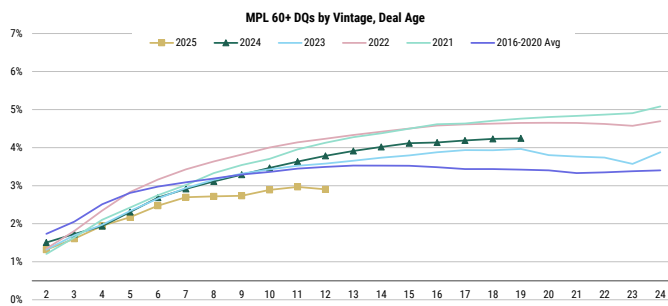
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 58: MPL FICO ≥ 660 60+DQs, annual comparison

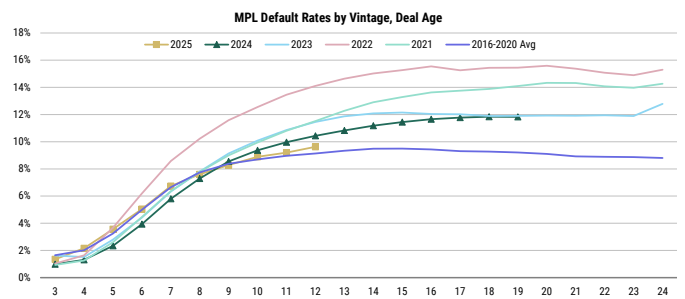
Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 59: MPL FICO ≥ 660 defaults, annual comparison

Source: Intex, Morgan Stanley Research. Deals are categorized by reported WAVG FICO at origination. "NO Fico" includes deals for which this metric could not be found. "Total" includes all MPL deals.

Exhibit 60: MPL 60+ DQs by Vintage, Deal Age

Source: Intex, Morgan Stanley Research

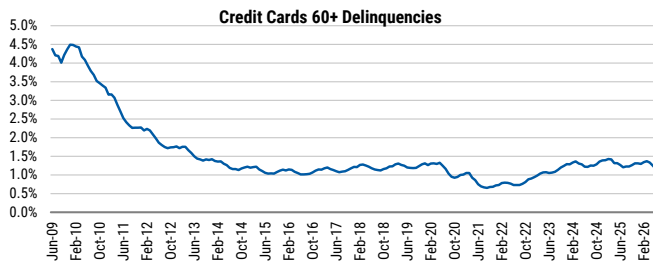
Exhibit 61: MPL 3mo CDR by Vintage, Deal Age

Source: Intex, Morgan Stanley Research

Performance of Credit Card Loans

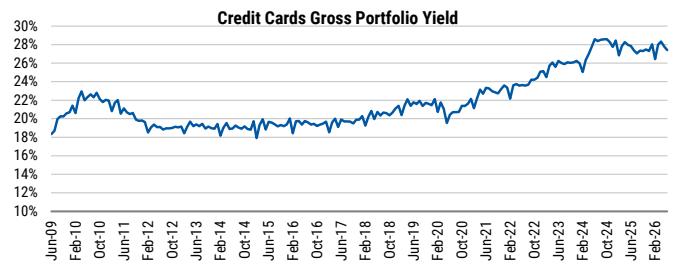
Credit card ABS 60+ delinquencies declined 9bps m/m to 1.17%. Charge-offs also declined in June, down 7bp to 3.01%. The monthly payment rate remained stable in June, decreasing 2bp to 41.53% from 41.55% in May.

Exhibit 62: 60+ delinquencies



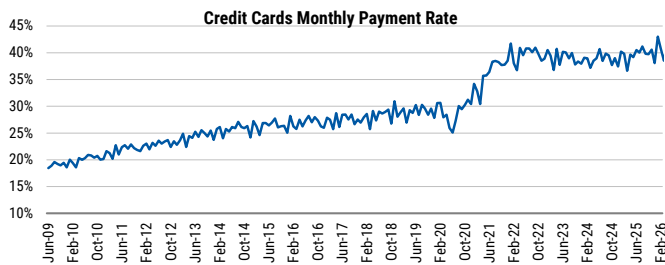
Source: Intex, Morgan Stanley Research

Exhibit 63: Gross portfolio yield



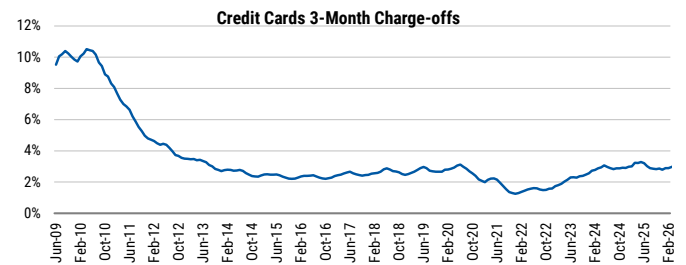
Source: Intex, Morgan Stanley Research

Exhibit 64: Monthly payment rate



Source: Intex, Morgan Stanley Research

Exhibit 65: 3-month charge-offs

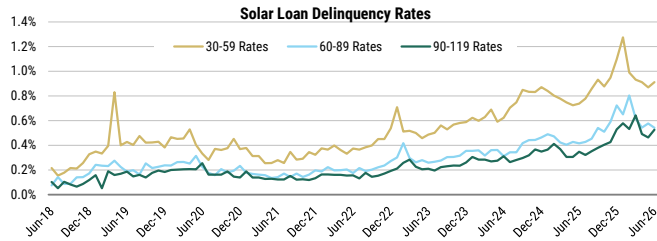


Source: Intex, Morgan Stanley Research

Solar Loan ABS

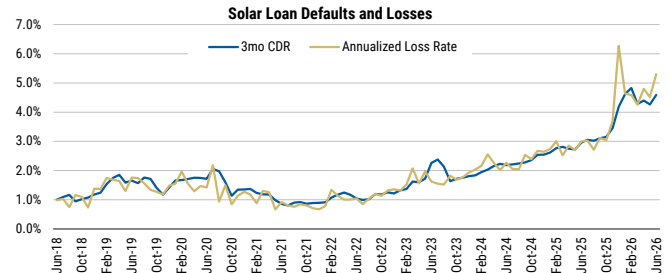
30-day delinquencies ticked up 4bp to 0.91%, returning to the level seen in April. Defaults and losses continue to climb, with 3mo CDR increasing 33bp to 4.59% from 4.26%. The 2024 and 2025 vintages continue to underperform the prior vintages. Annual net losses rose to 5.30%, up from 4.52% in May.

Exhibit 66: Delinquency rates on solar loan ABS



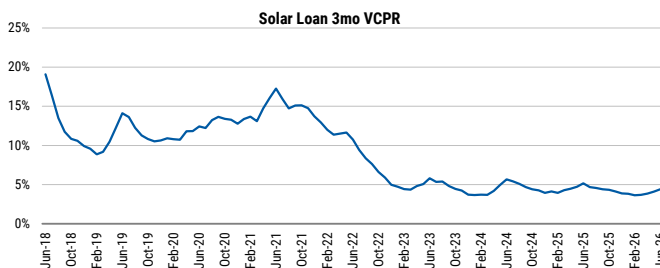
Source: Intex, Morgan Stanley Research

Exhibit 67: 3-month CDR and monthly annualized net loss rates



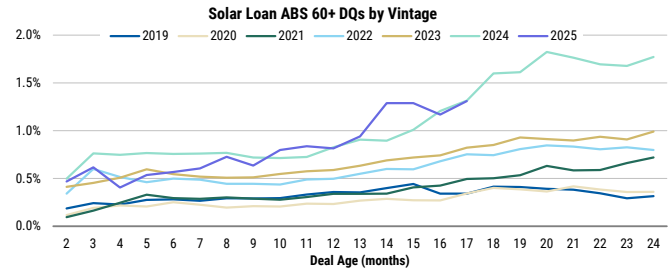
Source: Intex, Morgan Stanley Research

Exhibit 68: 3-month VCPR



Source: Intex, Morgan Stanley Research

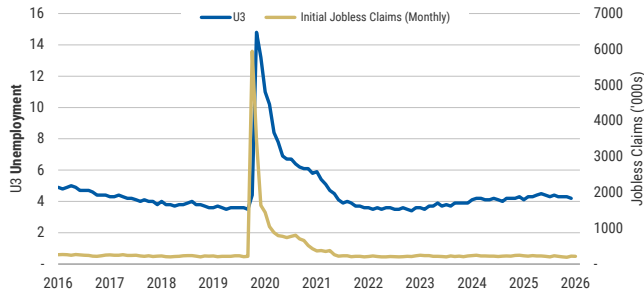
Exhibit 69: 60+ delinquencies by vintage and deal age



Source: Intex, Morgan Stanley Research

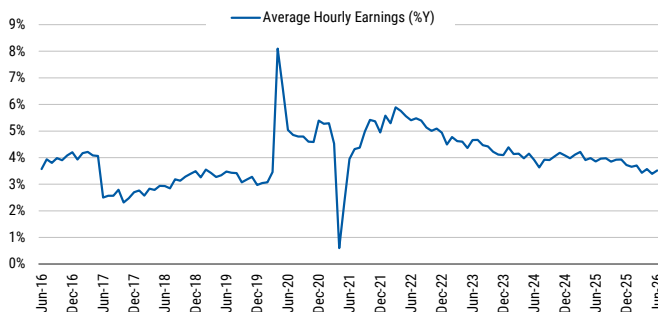
US Consumer Health Screen

Exhibit 70: U3 slightly decreased in June to 4.2% (-0.1%); recent jobless claims at 215k (-8k)



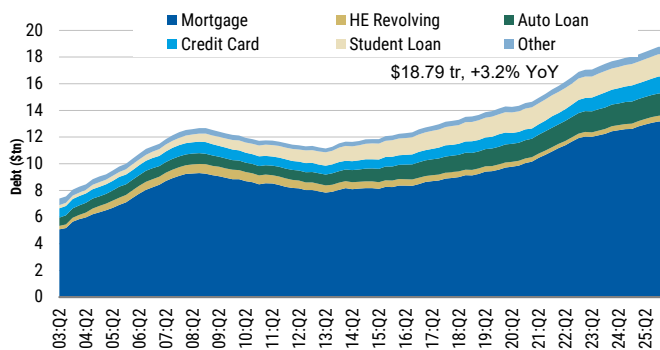
Source: BLS, Department of Labor, Morgan Stanley Research

Exhibit 72: AHE increased to 3.52% in June, up from 3.39% in May



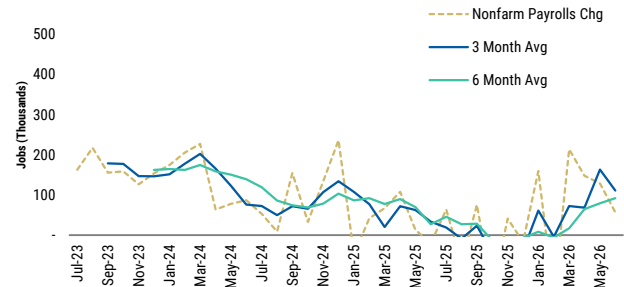
Source: BLS, Morgan Stanley Research

Exhibit 74: Household debt moved higher again in 1Q26



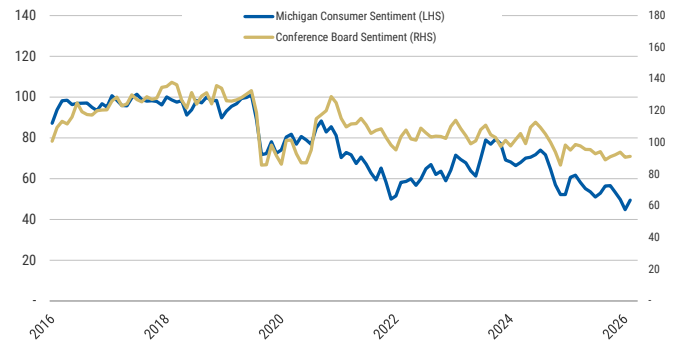
Source: NY Fed, Morgan Stanley Research

Exhibit 71: Payroll data showed a 57k increase in June



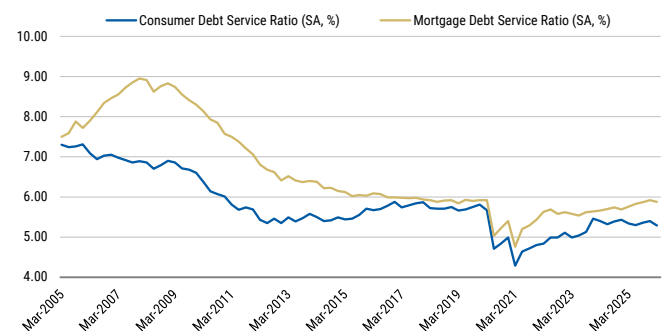
Source: BLS, Department of Labor, Morgan Stanley Research

Exhibit 73: June Michigan Consumer Sentiment increased to 49.5 following May's record low 44.8 (+0.7). June Conference Board Sentiment also increased m/m to 91.2 (+0.6).



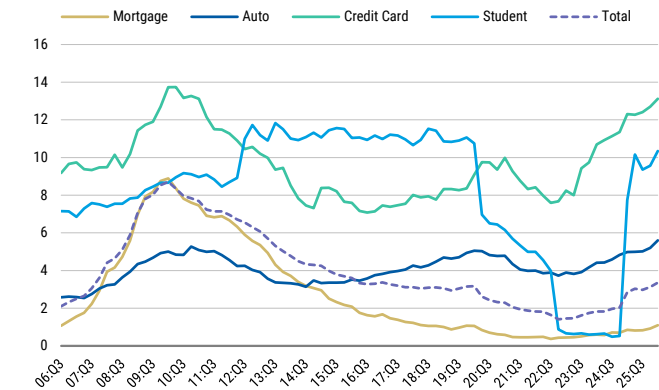
Source: University of Michigan, Bloomberg, Morgan Stanley Research

Exhibit 75: Both non-mortgage DTI and mortgage DTI crept lower through 1Q26



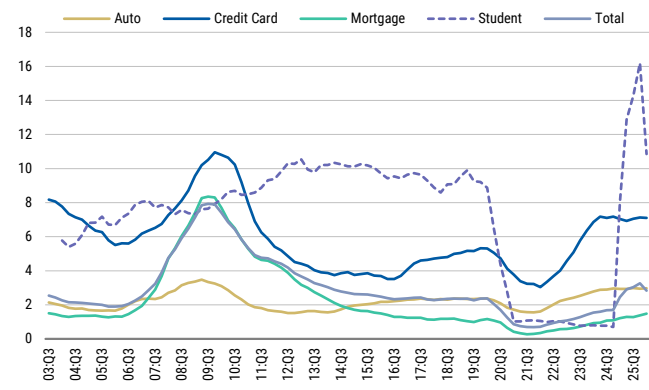
Source: Federal Reserve, Morgan Stanley Research

Exhibit 76: 90+ DQ balances saw an overall increase in 1Q26, with the sharpest uptick in student loans



Source: NY Fed, Morgan Stanley Research

Exhibit 77: Transition into 90+ decreased in 1Q26, with a pronounced decrease in student loans



Source: NY Fed, Morgan Stanley Research

Performance by Shelf

Exhibit 78: Subprime auto ABS: 3-mo CDR

month	ACAR	AMCAR	BLAST	CPS	DRIVE	EART	GCAR	RPM	SDART	WLAKE
Jan-22	18.19%	3.98%		8.95%	10.86%	11.73%	7.57%	8.33%	6.48%	6.62%
Feb-22	18.11%	3.72%		8.06%	10.11%	11.58%	8.10%	9.75%	7.42%	5.42%
Mar-22	19.69%	4.02%		8.67%	11.43%	12.79%	7.37%	6.83%	8.08%	6.28%
Apr-22	21.30%	4.37%		9.16%	12.08%	13.70%	8.66%	7.38%	8.21%	8.07%
May-22	18.63%	4.11%		8.35%	12.22%	11.92%	9.30%	7.97%	6.94%	8.50%
Jun-22	19.21%	3.32%		8.90%	11.99%	11.74%	8.25%	8.47%	5.95%	6.90%
Jul-22	20.19%	3.21%		7.86%	12.51%	10.23%	8.41%	9.37%	6.56%	6.67%
Aug-22	19.91%	3.69%		8.57%	13.81%	11.73%	9.52%	7.69%	6.80%	7.53%
Sep-22	23.88%	3.53%		9.55%	15.33%	12.78%	8.82%	9.17%	8.29%	7.52%
Oct-22	26.10%	4.06%		8.91%	16.52%	15.08%	10.44%	10.79%	8.59%	8.59%
Nov-22	24.55%	4.42%		9.97%	17.21%	16.29%	11.60%	8.10%	8.54%	9.54%
Dec-22	23.49%	4.76%		10.99%	17.12%	17.96%	11.57%	7.28%	8.45%	9.98%
Jan-23	23.20%	5.14%		9.84%	17.27%	17.81%	11.96%	8.42%	9.52%	8.97%
Feb-23	21.15%	5.31%		10.16%	17.25%	18.65%	13.03%	9.12%	9.96%	9.81%
Mar-23	21.43%	5.20%		11.12%	17.18%	17.18%	13.23%	10.02%	10.63%	10.62%
Apr-23	22.83%	5.37%		10.55%	15.80%	17.83%	13.70%	10.42%	9.75%	9.39%
May-23	19.75%	5.26%		11.77%	14.23%	15.66%	12.53%	11.03%	9.44%	9.72%
Jun-23	20.50%	4.38%		12.41%	12.85%	15.33%	11.39%	11.75%	9.28%	8.90%
Jul-23	21.06%	4.35%		11.33%	12.64%	15.01%	11.47%	10.08%	9.53%	9.46%
Aug-23	19.94%	4.62%		11.80%	13.24%	14.60%	11.37%	11.24%	9.42%	10.22%
Sep-23	22.06%	5.00%		12.11%	14.49%	14.82%	12.85%	15.06%	10.48%	11.11%
Oct-23	23.50%	5.58%		11.00%	15.31%	16.01%	13.71%	13.62%	10.17%	12.00%
Nov-23	21.25%	5.90%		11.76%	15.52%	15.99%	12.84%	17.51%	9.72%	10.80%
Dec-23	21.89%	5.06%		12.39%	15.31%	16.72%	13.18%	16.95%	9.79%	11.56%
Jan-24	23.42%	5.26%	3.50%	11.94%	15.22%	16.83%	13.80%	16.86%	9.35%	11.89%
Feb-24	20.45%	5.75%	7.50%	12.41%	15.61%	16.43%	12.61%	13.14%	9.44%	10.19%
Mar-24	21.75%	6.10%	12.15%	13.30%	15.75%	16.69%	13.36%	11.76%	9.97%	10.32%
Apr-24	22.77%	6.20%	9.95%	12.48%	15.37%	15.83%	13.89%	12.77%	9.44%	10.75%
May-24	20.23%	6.21%	12.54%	12.87%	7.55%	14.94%	11.84%	12.58%	9.57%	11.39%
Jun-24	20.78%	6.20%	14.95%	12.87%	8.10%	14.98%	11.82%	12.64%	9.87%	9.97%
Jul-24	21.15%	6.16%	11.58%	11.25%	9.00%	13.66%	11.85%	11.76%	9.32%	10.22%
Aug-24	18.48%	4.98%	14.26%	11.43%	10.68%	13.29%	10.64%	12.68%	10.04%	10.46%
Sep-24	21.22%	5.58%	17.29%	10.25%	12.80%	14.43%	12.82%	13.96%	9.71%	9.93%
Oct-24	22.64%	6.07%	14.01%	11.12%	14.91%	13.79%	14.23%	14.74%	10.48%	10.81%
Nov-24	20.39%	6.87%	15.58%	11.86%	15.96%	14.94%	13.15%	14.37%	9.96%	12.14%
Dec-24	20.84%	7.16%	16.99%	11.04%	8.40%	13.92%	12.84%	14.49%	10.58%	12.06%
Jan-25	21.97%	7.83%	14.42%	11.24%	8.97%	14.47%	13.20%	14.67%	9.98%	10.05%
Feb-25	19.72%	7.96%	15.88%	11.97%	10.20%	14.98%	11.82%	14.72%	10.65%	10.26%
Mar-25	20.68%	7.88%	16.36%	12.54%	11.32%	15.58%	12.34%	14.56%	10.99%	10.74%
Apr-25	21.80%	7.76%	14.26%	11.45%	12.27%	14.02%	13.01%	13.49%	9.92%	9.84%
May-25	17.79%	7.59%	14.86%	11.77%	12.65%	14.26%	10.75%	13.28%	9.71%	10.37%
Jun-25	18.57%	7.92%	16.72%	12.13%	13.19%	13.09%	10.97%	13.25%	8.86%	10.98%
Jul-25	19.58%	7.96%	17.38%	12.98%	13.74%	13.60%	10.97%	11.67%	9.26%	11.21%
Aug-25	18.89%	8.05%	15.63%	11.89%	9.80%	13.08%	9.89%	12.00%	10.10%	11.49%
Sep-25	20.68%	8.19%	17.25%	12.36%	10.86%	14.35%	10.69%	11.73%	11.04%	11.76%
Oct-25	21.94%	8.47%	16.60%	10.99%	11.98%	15.13%	11.84%	11.07%	10.68%	10.53%
Nov-25	20.52%	9.05%	17.88%	11.45%	13.25%	14.11%	11.34%	12.32%	11.35%	11.67%
Dec-25	21.08%	9.06%	18.60%	12.14%	9.86%	14.29%	12.11%	12.95%	11.87%	12.57%
Jan-26	22.52%	9.76%	17.26%	11.62%	11.32%	14.99%	12.65%	11.90%	12.24%	11.37%
Feb-26	20.11%	7.33%	18.50%	12.24%	12.54%	13.58%	11.25%	12.93%	11.34%	11.20%
Mar-26	21.58%	7.64%	18.31%	12.73%	13.79%	14.66%	11.86%	11.74%	11.65%	11.91%
Apr-26	22.57%	7.62%	15.79%	12.11%	14.21%	13.96%	12.87%	12.92%	11.60%	10.15%
May-26	18.29%	7.93%	15.48%	12.86%	13.99%	15.05%	11.03%	11.05%	10.04%	11.12%
Jun-26	19.44%	6.26%	17.20%	13.41%	13.51%	14.51%	11.15%	11.57%	9.93%	11.32%

Source: Intex, Morgan Stanley Research

Exhibit 79: Subprime auto ABS: 60+ delinquencies

month	ACAR	AMCAR	BLAST	CPS	DRIVE	EART	GCAR	RPM	SDART	WLAK
Jan-22	11.35%	1.52%		4.23%	4.97%	7.10%	4.23%	5.32%	3.41%	0.81%
Feb-22	12.87%	1.72%		3.89%	5.50%	7.89%	4.78%	4.03%	3.62%	1.10%
Mar-22	10.32%	1.53%		3.90%	5.65%	6.32%	4.36%	4.06%	3.20%	1.13%
Apr-22	9.45%	1.02%		3.77%	5.01%	5.52%	3.27%	4.04%	2.53%	0.62%
May-22	10.51%	1.14%		3.19%	4.75%	5.41%	3.44%	4.52%	2.53%	0.63%
Jun-22	10.44%	1.23%		3.68%	5.90%	6.58%	4.66%	5.54%	3.00%	0.88%
Jul-22	11.83%	1.27%		4.10%	6.73%	6.90%	4.23%	4.67%	3.74%	0.84%
Aug-22	12.23%	1.46%		3.92%	7.29%	8.12%	4.89%	5.40%	3.85%	1.02%
Sep-22	10.59%	1.76%		4.42%	7.56%	8.59%	5.70%	4.29%	3.79%	1.13%
Oct-22	9.43%	1.60%		4.74%	7.59%	9.03%	5.01%	4.64%	3.78%	1.18%
Nov-22	9.46%	1.82%		4.58%	7.55%	8.57%	5.46%	5.22%	4.17%	1.07%
Dec-22	9.36%	2.06%		5.17%	8.01%	9.79%	6.22%	4.92%	4.48%	1.13%
Jan-23	9.90%	2.08%		5.85%	8.80%	9.56%	6.51%	5.40%	5.17%	1.33%
Feb-23	11.41%	2.23%		5.44%	8.15%	10.02%	6.60%	6.19%	4.76%	1.15%
Mar-23	8.80%	1.82%		5.27%	7.34%	8.26%	5.72%	5.96%	4.61%	1.08%
Apr-23	7.65%	1.19%		4.96%	6.10%	7.09%	4.51%	5.74%	4.11%	0.91%
May-23	9.13%	1.58%		4.62%	6.03%	7.85%	4.89%	6.58%	4.16%	1.02%
Jun-23	8.30%	1.57%		5.06%	6.93%	7.81%	5.89%	5.85%	4.37%	1.03%
Jul-23	8.58%	1.61%		5.49%	7.69%	7.63%	6.24%	6.94%	4.92%	1.17%
Aug-23	9.33%	1.90%		5.56%	8.53%	8.40%	6.86%	6.59%	5.14%	1.31%
Sep-23	8.42%	2.09%		6.22%	8.74%	8.29%	6.35%	6.48%	5.07%	1.14%
Oct-23	8.49%	1.65%		6.88%	8.55%	8.42%	6.56%	6.57%	5.18%	1.21%
Nov-23	8.62%	1.94%	0.04%	6.63%	8.94%	8.78%	6.95%	6.13%	5.17%	1.24%
Dec-23	7.75%	2.18%	0.74%	7.20%	9.41%	8.45%	6.38%	5.88%	5.27%	1.18%
Jan-24	8.69%	2.23%	1.87%	7.92%	10.12%	9.15%	6.65%	6.31%	5.80%	1.32%
Feb-24	9.30%	2.26%	2.14%	7.39%	10.20%	8.52%	7.10%	5.87%	5.67%	1.49%
Mar-24	6.95%	2.05%	2.71%	7.16%	4.77%	7.53%	5.37%	5.39%	5.39%	1.56%
Apr-24	6.21%	1.87%	3.46%	7.15%	4.89%	7.12%	4.98%	5.29%	5.22%	1.26%
May-24	6.94%	1.97%	3.48%	6.51%	5.52%	6.88%	6.10%	5.27%	4.89%	1.27%
Jun-24	6.14%	1.61%	4.33%	6.80%	6.21%	6.64%	5.41%	5.76%	5.19%	1.33%
Jul-24	7.32%	1.88%	5.48%	6.44%	7.12%	7.60%	5.78%	6.51%	5.21%	1.31%
Aug-24	8.14%	2.17%	4.70%	7.07%	7.60%	7.15%	6.63%	7.24%	5.62%	1.48%
Sep-24	6.74%	2.38%	4.80%	7.53%	7.74%	7.75%	5.64%	7.00%	5.18%	1.62%
Oct-24	7.34%	2.53%	5.53%	8.24%	4.31%	7.38%	6.07%	7.31%	5.70%	1.68%
Nov-24	7.55%	2.54%	5.07%	7.49%	4.94%	7.03%	6.30%	7.24%	5.44%	1.28%
Dec-24	6.87%	2.75%	5.44%	7.99%	5.91%	7.83%	5.54%	7.49%	5.76%	1.31%
Jan-25	8.53%	3.13%	6.78%	8.76%	6.71%	9.04%	6.41%	8.04%	6.29%	1.56%
Feb-25	9.07%	3.05%	5.67%	8.82%	7.00%	7.96%	6.32%	7.61%	5.87%	1.27%
Mar-25	6.70%	2.72%	5.57%	7.50%	6.98%	7.46%	4.78%	7.06%	5.71%	1.36%
Apr-25	6.11%	2.46%	5.82%	7.59%	6.86%	6.51%	4.61%	6.84%	5.03%	1.40%
May-25	7.19%	2.70%	7.22%	7.80%	6.91%	7.29%	5.43%	6.43%	5.07%	1.23%
Jun-25	6.41%	2.57%	5.87%	7.01%	5.30%	6.95%	4.75%	6.56%	5.91%	1.20%
Jul-25	7.30%	2.97%	6.56%	7.59%	6.22%	7.61%	5.32%	6.45%	6.70%	1.33%
Aug-25	7.99%	3.05%	6.52%	7.04%	6.92%	8.18%	5.48%	7.01%	6.22%	1.11%
Sep-25	7.35%	3.35%	6.98%	7.63%	7.57%	7.71%	4.92%	6.90%	6.49%	1.24%
Oct-25	7.62%	3.60%	8.14%	7.92%	5.74%	7.72%	5.54%	7.39%	7.01%	1.35%
Nov-25	7.52%	3.51%	7.24%	7.03%	5.94%	7.58%	5.51%	6.60%	6.96%	1.16%
Dec-25	7.88%	3.18%	7.91%	7.65%	7.23%	7.49%	5.02%	7.74%	6.89%	1.27%
Jan-26	9.37%	3.36%	9.16%	8.07%	8.15%	8.21%	5.94%	7.38%	7.38%	1.24%
Feb-26	9.96%	3.19%	7.33%	7.47%	8.34%	8.27%	6.01%	7.82%	7.41%	1.13%
Mar-26	7.24%	2.99%	6.67%	7.07%	7.94%	7.84%	4.49%	6.07%	6.46%	1.13%
Apr-26	7.19%	2.18%	7.29%	6.75%	7.44%	7.00%	4.41%	6.29%	6.18%	1.13%
May-26	8.20%	2.37%	6.80%	6.61%	7.08%	7.23%	5.16%	6.38%	5.96%	1.11%
Jun-26	6.84%	2.46%	7.07%	6.09%	8.17%	7.88%	4.40%	6.19%	6.95%	1.02%

Source: Intex, Morgan Stanley Research

Exhibit 80: Prime auto ABS: 3-mo CDR

month	CARMX	FORDO	FORDR	GMCAR	GMREV	HAROT	HART	TALNT	TAOT	WOART
Jan-22	1.54%	0.54%	0.50%	0.41%	0.28%	0.18%	1.00%	0.50%	0.39%	0.83%
Feb-22	1.79%	0.60%	0.50%	0.41%	0.46%	0.16%	0.85%	0.57%	0.37%	0.84%
Mar-22	1.77%	0.63%	0.47%	0.46%	0.54%	0.17%	0.93%	0.57%	0.38%	0.85%
Apr-22	1.51%	0.54%	0.46%	0.44%	0.55%	0.18%	1.00%	0.60%	0.36%	0.96%
May-22	1.27%	0.56%	0.44%	0.43%	0.47%	0.15%	0.90%	0.51%	0.35%	0.81%
Jun-22	1.33%	0.62%	0.44%	0.43%	0.43%	0.16%	0.68%	0.48%	0.36%	0.85%
Jul-22	1.30%	0.65%	0.42%	0.36%	0.50%	0.18%	0.73%	0.43%	0.31%	0.86%
Aug-22	1.58%	0.69%	0.42%	0.42%	0.57%	0.20%	0.90%	0.40%	0.33%	0.88%
Sep-22	1.86%	0.58%	0.45%	0.46%	0.60%	0.23%	1.07%	0.50%	0.39%	0.87%
Oct-22	1.76%	0.57%	0.50%	0.44%	0.64%	0.23%	0.98%	0.56%	0.42%	0.91%
Nov-22	1.92%	0.58%	0.51%	0.48%	0.69%	0.24%	1.15%	0.65%	0.42%	0.94%
Dec-22	2.07%	0.42%	0.50%	0.49%	0.71%	0.24%	1.21%	0.70%	0.46%	1.00%
Jan-23	2.06%	0.48%	0.50%	0.44%	0.70%	0.28%	1.22%	0.78%	0.53%	1.07%
Feb-23	2.26%	0.46%	0.57%	0.47%	0.52%	0.29%	1.00%	0.88%	0.56%	0.95%
Mar-23	2.22%	0.49%	0.64%	0.50%	0.57%	0.30%	1.08%	0.92%	0.61%	0.99%
Apr-23	1.90%	0.58%	0.68%	0.49%	0.63%	0.30%	1.10%	0.94%	0.57%	1.12%
May-23	1.80%	0.61%	0.55%	0.52%	0.58%	0.24%	1.07%	0.83%	0.57%	0.96%
Jun-23	1.89%	0.51%	0.54%	0.57%	0.53%	0.24%	1.10%	0.77%	0.58%	0.98%
Jul-23	1.80%	0.55%	0.58%	0.51%	0.54%	0.25%	0.96%	0.74%	0.56%	0.89%
Aug-23	2.17%	0.66%	0.60%	0.52%	0.58%	0.23%	1.06%	0.78%	0.51%	0.95%
Sep-23	2.47%	0.65%	0.65%	0.57%	0.47%	0.29%	1.12%	0.81%	0.56%	1.06%
Oct-23	2.29%	0.70%	0.65%	0.55%	0.50%	0.37%	0.98%	0.75%	0.63%	1.12%
Nov-23	2.37%	0.74%	0.59%	0.61%	0.56%	0.33%	1.10%	0.79%	0.60%	1.10%
Dec-23	2.61%	0.86%	0.64%	0.62%	0.64%	0.35%	1.19%	0.85%	0.66%	1.26%
Jan-24	2.53%	0.89%	0.71%	0.65%	0.63%	0.35%	1.27%	0.90%	0.73%	1.32%
Feb-24	2.83%	0.81%	0.79%	0.68%	0.56%	0.38%	1.07%	1.01%	0.73%	1.24%
Mar-24	2.92%	0.82%	0.78%	0.75%	0.62%	0.43%	1.12%	1.10%	0.81%	1.35%
Apr-24	2.62%	0.99%	0.81%	0.63%	0.79%	0.48%	1.21%	1.11%	0.76%	1.51%
May-24	2.53%	0.99%	0.71%	0.68%	0.79%	0.34%	1.27%	1.09%	0.77%	1.39%
Jun-24	2.50%	0.90%	0.77%	0.72%	0.61%	0.38%	1.11%	1.00%	0.77%	1.44%
Jul-24	2.35%	0.84%	0.76%	0.71%	0.67%	0.41%	1.13%	0.99%	0.72%	1.44%
Aug-24	2.69%	1.06%	0.84%	0.77%	0.72%	0.37%	1.19%	1.04%	0.79%	1.29%
Sep-24	3.09%	0.91%	0.83%	0.76%	0.74%	0.42%	1.26%	1.08%	0.94%	1.46%
Oct-24	2.85%	0.99%	0.85%	0.72%	0.74%	0.45%	1.16%	1.17%	0.88%	1.61%
Nov-24	2.94%	1.05%	0.86%	0.83%	0.81%	0.44%	1.29%	1.23%	0.96%	1.71%
Dec-24	2.90%	0.92%	0.92%	0.92%	0.82%	0.49%	1.37%	1.14%	0.97%	1.70%
Jan-25	2.54%	1.04%	0.95%	0.89%	0.87%	0.51%	1.27%	1.11%	0.89%	1.87%
Feb-25	2.54%	0.91%	0.94%	0.90%	0.93%	0.63%	1.35%	1.10%	0.92%	1.98%
Mar-25	2.60%	0.95%	0.93%	0.93%	0.86%	0.68%	1.44%	1.10%	0.93%	2.10%
Apr-25	2.43%	1.03%	1.00%	0.88%	0.94%	0.71%	1.39%	1.10%	0.83%	1.75%
May-25	2.45%	1.07%	0.84%	0.99%	1.02%	0.58%	1.30%	1.01%	0.84%	1.62%
Jun-25	2.51%	0.96%	0.88%	1.05%	0.86%	0.61%	1.08%	1.01%	0.89%	1.74%
Jul-25	2.39%	0.97%	0.89%	1.06%	0.90%	0.61%	1.18%	1.00%	0.81%	1.84%
Aug-25	2.61%	1.04%	0.93%	0.95%	0.87%	0.54%	1.36%	1.02%	0.84%	1.69%
Sep-25	2.86%	1.11%	1.01%	1.01%	0.98%	0.64%	1.37%	0.86%	0.92%	1.75%
Oct-25	2.58%	1.27%	1.05%	0.97%	1.04%	0.74%	1.51%	0.94%	0.84%	1.92%
Nov-25	2.81%	1.40%	0.94%	1.09%	1.15%	0.69%	1.65%	1.02%	0.94%	1.89%
Dec-25	2.88%	1.16%	0.91%	1.09%	1.08%	0.72%	1.40%	1.07%	0.98%	1.97%
Jan-26	2.62%	1.22%	0.95%	1.17%	1.08%	0.73%	1.52%	1.10%	0.97%	1.91%
Feb-26	2.68%	1.08%	1.04%	1.19%	1.11%	0.71%	1.39%	1.05%	1.05%	2.16%
Mar-26	2.57%	1.18%	1.06%	1.33%	1.17%	0.76%	1.44%	1.04%	1.08%	2.51%
Apr-26	2.32%	1.19%	1.09%	1.13%	1.24%	0.82%	1.41%	1.09%	0.97%	2.63%
May-26	2.18%	1.32%	0.90%	1.15%	1.30%	0.68%	1.18%	1.01%	1.01%	2.26%
Jun-26	2.27%	1.13%	0.97%	1.15%	1.34%	0.69%	1.20%	1.05%	1.06%	2.16%

Source: Intex, Morgan Stanley Research

Exhibit 81: Prime auto ABS: 60+ delinquencies

month	CARMX	FORDO	FORDR	GMCAR	GMREV	HAROT	HART	TALNT	TAOT	WOART
Jan-22	1.44%	0.18%	0.14%	0.14%	0.12%	0.11%	0.33%	0.26%	0.22%	0.25%
Feb-22	1.30%	0.22%	0.15%	0.15%	0.13%	0.13%	0.36%	0.29%	0.21%	0.29%
Mar-22	1.26%	0.14%	0.13%	0.14%	0.12%	0.12%	0.31%	0.28%	0.22%	0.26%
Apr-22	1.11%	0.13%	0.12%	0.10%	0.11%	0.09%	0.20%	0.21%	0.16%	0.22%
May-22	1.14%	0.16%	0.14%	0.12%	0.13%	0.10%	0.26%	0.22%	0.17%	0.26%
Jun-22	1.29%	0.20%	0.15%	0.11%	0.14%	0.12%	0.33%	0.21%	0.18%	0.28%
Jul-22	1.48%	0.16%	0.14%	0.13%	0.11%	0.13%	0.41%	0.23%	0.20%	0.28%
Aug-22	1.42%	0.15%	0.13%	0.13%	0.14%	0.15%	0.39%	0.27%	0.22%	0.36%
Sep-22	1.60%	0.18%	0.15%	0.16%	0.19%	0.13%	0.46%	0.29%	0.23%	0.33%
Oct-22	1.60%	0.16%	0.14%	0.14%	0.24%	0.13%	0.40%	0.30%	0.24%	0.32%
Nov-22	1.53%	0.15%	0.15%	0.13%	0.22%	0.15%	0.39%	0.33%	0.26%	0.36%
Dec-22	1.74%	0.15%	0.18%	0.17%	0.15%	0.16%	0.38%	0.40%	0.27%	0.35%
Jan-23	1.82%	0.16%	0.20%	0.17%	0.17%	0.17%	0.44%	0.43%	0.31%	0.38%
Feb-23	1.73%	0.19%	0.20%	0.16%	0.16%	0.18%	0.44%	0.45%	0.32%	0.38%
Mar-23	1.61%	0.17%	0.16%	0.15%	0.14%	0.12%	0.41%	0.40%	0.31%	0.31%
Apr-23	1.48%	0.12%	0.12%	0.13%	0.13%	0.10%	0.34%	0.33%	0.26%	0.28%
May-23	1.51%	0.13%	0.15%	0.14%	0.14%	0.12%	0.34%	0.39%	0.30%	0.29%
Jun-23	1.71%	0.18%	0.18%	0.15%	0.15%	0.13%	0.34%	0.38%	0.27%	0.34%
Jul-23	1.80%	0.14%	0.18%	0.15%	0.11%	0.14%	0.35%	0.40%	0.29%	0.35%
Aug-23	1.73%	0.17%	0.19%	0.15%	0.15%	0.17%	0.33%	0.38%	0.32%	0.39%
Sep-23	1.92%	0.20%	0.18%	0.17%	0.13%	0.14%	0.37%	0.42%	0.31%	0.40%
Oct-23	1.97%	0.20%	0.19%	0.18%	0.18%	0.15%	0.39%	0.41%	0.33%	0.43%
Nov-23	1.89%	0.22%	0.19%	0.17%	0.16%	0.17%	0.40%	0.44%	0.35%	0.43%
Dec-23	2.07%	0.22%	0.21%	0.21%	0.20%	0.17%	0.38%	0.48%	0.36%	0.41%
Jan-24	2.28%	0.26%	0.24%	0.21%	0.18%	0.19%	0.44%	0.53%	0.42%	0.49%
Feb-24	2.07%	0.29%	0.24%	0.18%	0.20%	0.19%	0.46%	0.58%	0.41%	0.52%
Mar-24	1.84%	0.25%	0.21%	0.19%	0.16%	0.13%	0.41%	0.51%	0.40%	0.43%
Apr-24	1.76%	0.25%	0.17%	0.18%	0.14%	0.11%	0.32%	0.46%	0.37%	0.42%
May-24	1.74%	0.23%	0.20%	0.19%	0.11%	0.13%	0.37%	0.45%	0.33%	0.41%
Jun-24	1.84%	0.25%	0.20%	0.18%	0.13%	0.13%	0.38%	0.48%	0.36%	0.39%
Jul-24	2.02%	0.25%	0.18%	0.20%	0.14%	0.14%	0.41%	0.51%	0.44%	0.49%
Aug-24	1.90%	0.24%	0.21%	0.21%	0.19%	0.17%	0.38%	0.53%	0.42%	0.54%
Sep-24	1.97%	0.24%	0.23%	0.23%	0.19%	0.17%	0.39%	0.53%	0.45%	0.55%
Oct-24	1.78%	0.24%	0.25%	0.25%	0.19%	0.18%	0.43%	0.53%	0.47%	0.53%
Nov-24	1.35%	0.24%	0.24%	0.23%	0.19%	0.17%	0.42%	0.52%	0.41%	0.59%
Dec-24	1.41%	0.27%	0.24%	0.24%	0.22%	0.19%	0.49%	0.50%	0.42%	0.58%
Jan-25	1.63%	0.25%	0.29%	0.29%	0.24%	0.22%	0.53%	0.52%	0.41%	0.62%
Feb-25	1.72%	0.28%	0.26%	0.28%	0.26%	0.24%	0.44%	0.51%	0.38%	0.51%
Mar-25	1.37%	0.23%	0.21%	0.28%	0.26%	0.19%	0.37%	0.45%	0.37%	0.48%
Apr-25	1.39%	0.19%	0.19%	0.24%	0.18%	0.15%	0.29%	0.41%	0.34%	0.44%
May-25	1.45%	0.23%	0.21%	0.26%	0.20%	0.16%	0.32%	0.40%	0.33%	0.47%
Jun-25	1.57%	0.24%	0.21%	0.23%	0.20%	0.16%	0.35%	0.41%	0.34%	0.44%
Jul-25	1.71%	0.30%	0.23%	0.27%	0.20%	0.19%	0.37%	0.36%	0.37%	0.49%
Aug-25	1.53%	0.30%	0.22%	0.26%	0.23%	0.20%	0.39%	0.40%	0.36%	0.52%
Sep-25	1.70%	0.27%	0.21%	0.30%	0.25%	0.18%	0.43%	0.42%	0.40%	0.57%
Oct-25	1.66%	0.29%	0.22%	0.30%	0.27%	0.19%	0.40%	0.40%	0.40%	0.57%
Nov-25	1.44%	0.25%	0.21%	0.29%	0.26%	0.20%	0.42%	0.42%	0.38%	0.53%
Dec-25	1.53%	0.28%	0.22%	0.34%	0.32%	0.21%	0.45%	0.47%	0.45%	0.65%
Jan-26	1.61%	0.29%	0.26%	0.36%	0.33%	0.24%	0.52%	0.47%	0.46%	0.68%
Feb-26	1.47%	0.31%	0.28%	0.33%	0.36%	0.26%	0.50%	0.46%	0.43%	0.71%
Mar-26	1.38%	0.29%	0.24%	0.32%	0.33%	0.19%	0.37%	0.40%	0.41%	0.57%
Apr-26	1.28%	0.23%	0.19%	0.31%	0.26%	0.17%	0.34%	0.33%	0.39%	0.55%
May-26	1.27%	0.26%	0.20%	0.29%	0.28%	0.19%	0.35%	0.40%	0.37%	0.59%
Jun-26	1.34%	0.25%	0.22%	0.30%	0.28%	0.20%	0.37%	0.33%	0.40%	0.57%

Source: Intex, Morgan Stanley Research

Exhibit 82: MPL ABS: 3-mo CDR

month	AFFRM	AFRMT	LFT	MFIT	OMFIT	OPTN	PAID	SCLP	SLCLC	UPST
Jan-22	4.63%		4.15%	6.38%	5.13%	8.09%	5.46%	3.66%		4.68%
Feb-22	5.98%		4.90%	4.90%	5.60%	6.97%	6.96%	2.89%		6.17%
Mar-22	5.91%		5.38%	5.08%	5.95%	8.43%	6.58%	3.05%		6.05%
Apr-22	5.87%		5.77%	6.07%	6.20%	9.91%	7.26%	3.07%		7.80%
May-22	5.63%		6.41%	7.30%	6.24%	10.44%	8.22%	3.29%		9.24%
Jun-22	5.23%		6.99%	8.46%	6.30%	10.89%	7.61%	3.30%		10.76%
Jul-22	5.57%		7.56%	8.82%	5.57%	9.18%	8.89%	3.26%		9.70%
Aug-22	5.99%		7.52%	8.90%	5.50%	10.44%	9.79%	3.64%		9.89%
Sep-22	5.01%		6.23%	9.08%	5.41%	9.96%	9.57%	4.01%		9.07%
Oct-22	5.24%		6.23%	9.81%	5.32%	11.56%	10.81%	3.90%		9.79%
Nov-22	6.71%		6.68%	10.76%	5.61%	11.90%	10.23%	3.95%		12.31%
Dec-22	7.20%		6.89%	11.37%	4.85%	13.68%	11.47%	3.80%		13.72%
Jan-23	7.33%		7.88%	9.13%	5.40%	15.07%	13.02%	5.86%		16.21%
Feb-23	7.82%		8.97%	8.96%	5.69%	13.27%	11.89%	6.49%		18.26%
Mar-23	8.02%		10.33%	9.84%	5.02%	12.52%	12.78%	1.24%		18.78%
Apr-23	8.04%		11.44%	10.83%	5.37%	12.72%	13.19%	1.64%		18.95%
May-23	5.00%		12.26%	12.10%	5.74%	13.55%	13.71%	2.94%		18.72%
Jun-23	5.04%		12.25%	12.22%	6.08%	14.51%	12.70%	2.51%		17.10%
Jul-23	5.10%		9.59%	11.68%	6.18%	14.36%	13.13%	3.10%		18.43%
Aug-23	4.88%		9.37%	10.74%	5.47%	14.01%	12.80%	3.88%	0.02%	19.26%
Sep-23	5.45%		9.19%	10.09%	5.55%	13.36%	14.27%	4.85%	0.14%	20.48%
Oct-23	5.95%		8.88%	9.84%	5.67%	13.39%	13.96%	5.90%	0.52%	19.57%
Nov-23	6.79%		9.20%	9.30%	4.68%	14.11%	15.44%	6.72%	0.68%	20.37%
Dec-23	7.06%		9.55%	9.29%	5.18%	14.60%	14.38%	6.99%	0.96%	20.91%
Jan-24	4.82%		10.27%	7.65%	5.80%	14.83%	15.56%	7.47%	0.85%	21.13%
Feb-24	5.60%		10.77%	8.23%	6.41%	14.82%	14.86%	7.88%	0.90%	18.99%
Mar-24	5.70%		11.24%	8.98%	7.06%	14.73%	14.07%	8.35%	0.99%	19.11%
Apr-24	6.50%		9.57%	9.59%	7.76%	14.49%	13.73%	8.37%	1.24%	20.00%
May-24	6.53%		10.02%	10.54%	8.30%	12.66%	13.09%	8.40%	1.54%	19.51%
Jun-24	4.65%		9.93%	10.78%	8.58%	13.11%	12.58%	7.89%	2.06%	19.43%
Jul-24	4.91%		9.85%	10.58%	7.43%	13.19%	12.60%	7.84%	2.54%	18.72%
Aug-24	4.55%		9.70%	10.16%	7.22%	13.47%	11.55%	7.83%	2.96%	18.89%
Sep-24	5.29%		10.04%	9.75%	6.91%	14.01%	11.25%	8.81%	2.93%	19.52%
Oct-24	6.10%		10.24%	9.80%	6.57%	14.31%	11.46%	8.64%	3.36%	19.40%
Nov-24	7.48%		10.32%	6.97%	6.71%	15.05%	11.89%	8.75%	3.57%	19.48%
Dec-24	8.50%		10.35%	7.09%	6.98%	12.97%	11.80%	8.66%	3.78%	19.33%
Jan-25	7.38%		9.47%	8.11%	7.59%	11.36%	13.05%	9.19%	4.10%	19.65%
Feb-25	8.08%		9.57%	6.94%	7.82%	12.42%	13.09%	8.83%	4.43%	15.58%
Mar-25	7.26%		9.72%	7.92%	8.10%	13.07%	13.23%	7.84%	4.95%	15.10%
Apr-25	7.75%		9.55%	8.46%	8.31%	11.29%	12.28%	6.43%	5.20%	14.94%
May-25	7.73%		9.96%	9.54%	8.34%	11.68%	11.33%	7.09%	5.23%	15.09%
Jun-25	8.10%	3.04%	10.32%	9.99%	8.48%	12.11%	11.26%	0.76%	5.18%	15.70%
Jul-25	8.25%	5.17%	10.25%	10.19%	8.40%	13.34%	11.40%	2.54%	5.45%	15.90%
Aug-25	7.08%	5.22%	8.63%	8.18%	8.30%	14.27%	11.85%	2.18%	5.51%	11.02%
Sep-25	7.03%	5.31%	8.32%	7.92%	7.22%	10.69%	13.16%	3.69%	5.77%	9.17%
Oct-25	7.60%	5.36%	7.09%	7.82%	7.19%	11.22%	12.95%	3.28%	5.88%	9.73%
Nov-25	8.66%	5.80%	7.15%	8.06%	7.55%	12.64%	12.59%	4.63%	6.12%	11.96%
Dec-25	9.31%	6.10%	7.40%	9.05%	7.93%	10.68%	13.63%	4.65%	6.05%	9.81%
Jan-26	10.05%	6.62%	8.16%	8.15%	8.57%	11.67%	13.87%	4.45%	6.48%	11.20%
Feb-26	7.66%	7.72%	7.87%	8.61%	9.04%	10.37%	13.03%	5.46%	6.38%	9.66%
Mar-26	7.83%	8.13%	8.73%	8.99%	9.39%	11.15%	12.51%	6.39%	5.80%	10.20%
Apr-26	8.39%	7.24%	9.49%	9.38%	9.66%	12.49%	13.42%	7.30%	6.09%	12.32%
May-26	9.18%	6.30%	9.98%	10.07%	9.85%	10.18%	12.09%	5.08%	5.90%	12.84%
Jun-26	10.17%	6.39%	10.13%	10.37%	10.00%	11.03%	12.36%	5.63%	6.07%	11.70%

Source: Intex, Morgan Stanley Research

Exhibit 83: MPL ABS: 60+ delinquencies

month	AFFRM	AFRMT	LFT	MFIT	OMFIT	OPTN	PAID	SCLP	SLCLC	UPST
Jan-22	1.23%		3.10%	3.74%	2.53%	2.01%	1.50%	0.88%		1.38%
Feb-22	1.30%		3.39%	4.34%	2.61%	2.27%	1.86%	0.96%		1.77%
Mar-22	1.32%		3.53%	4.75%	2.65%	2.30%	1.81%	0.91%		2.05%
Apr-22	1.04%		3.68%	4.77%	2.62%	2.28%	2.13%	0.90%		2.15%
May-22	1.18%		3.64%	4.75%	2.31%	1.91%	2.66%	1.00%		2.01%
Jun-22	1.19%		3.80%	4.86%	2.35%	2.14%	2.18%	1.04%		2.04%
Jul-22	1.16%		3.04%	5.11%	2.53%	2.13%	2.35%	1.11%		1.89%
Aug-22	1.32%		3.39%	5.45%	2.66%	2.45%	2.44%	1.11%		2.24%
Sep-22	1.50%		3.74%	5.80%	2.82%	2.64%	2.74%	1.23%		2.80%
Oct-22	1.38%		4.26%	6.08%	2.77%	3.09%	3.14%	1.59%		3.03%
Nov-22	1.45%		4.85%	5.20%	2.39%	3.21%	2.89%	1.79%		3.55%
Dec-22	1.60%		5.59%	5.89%	2.60%	2.74%	2.99%	1.85%		3.94%
Jan-23	1.67%		6.17%	6.34%	2.33%	2.57%	3.22%	1.46%		4.12%
Feb-23	1.66%		6.47%	6.71%	2.50%	2.87%	3.11%	0.37%		4.15%
Mar-23	1.21%		6.41%	6.68%	2.57%	3.04%	2.90%	0.67%		4.07%
Apr-23	1.01%		6.33%	6.16%	2.52%	3.08%	2.58%	0.85%		3.58%
May-23	1.04%		4.93%	5.90%	2.53%	3.17%	2.87%	0.58%		4.01%
Jun-23	1.01%		4.57%	5.44%	2.29%	3.08%	3.10%	0.80%		4.27%
Jul-23	1.13%		4.67%	5.10%	2.33%	2.84%	3.28%	1.07%		4.18%
Aug-23	1.27%		4.79%	5.25%	2.54%	2.92%	3.32%	1.14%	0.04%	4.37%
Sep-23	1.39%		4.90%	5.45%	2.31%	3.33%	3.34%	1.33%	0.11%	4.22%
Oct-23	1.42%		5.23%	5.63%	2.51%	3.21%	3.53%	1.42%	0.26%	4.25%
Nov-23	1.06%		5.42%	5.04%	2.70%	3.26%	3.30%	1.49%	0.37%	4.44%
Dec-23	1.36%		5.74%	5.48%	3.01%	3.57%	3.27%	1.78%	0.35%	4.14%
Jan-24	1.22%		6.00%	5.91%	3.27%	3.34%	3.53%	1.79%	0.34%	4.27%
Feb-24	1.34%		5.01%	6.23%	3.51%	3.45%	3.67%	1.77%	0.34%	4.58%
Mar-24	1.43%		4.96%	6.15%	3.56%	3.00%	3.08%	1.81%	0.41%	4.39%
Apr-24	1.17%		5.10%	5.95%	3.37%	2.84%	2.91%	1.69%	0.41%	3.82%
May-24	1.16%		5.02%	5.77%	2.86%	2.91%	3.09%	1.61%	0.52%	4.16%
Jun-24	0.90%		5.06%	5.58%	2.68%	2.95%	3.04%	1.82%	0.61%	4.25%
Jul-24	1.03%		5.27%	5.66%	2.68%	2.96%	2.78%	2.03%	0.60%	4.15%
Aug-24	1.21%		5.30%	5.77%	2.83%	3.27%	2.98%	1.92%	0.67%	4.27%
Sep-24	1.50%		5.39%	4.64%	2.96%	3.37%	2.97%	1.94%	0.70%	4.31%
Oct-24	1.73%		5.52%	4.78%	3.18%	2.97%	2.95%	2.12%	0.71%	4.29%
Nov-24	1.51%		4.97%	4.26%	3.29%	2.57%	2.98%	2.08%	0.79%	4.38%
Dec-24	1.68%		5.14%	4.72%	3.43%	2.76%	3.05%	2.09%	0.89%	3.55%
Jan-25	1.49%		5.51%	5.09%	3.56%	3.11%	3.25%	1.95%	0.90%	3.52%
Feb-25	1.48%		5.57%	5.38%	3.56%	2.60%	3.03%	1.93%	0.94%	3.54%
Mar-25	1.52%		5.50%	5.45%	3.56%	2.79%	2.78%	1.98%	0.88%	3.39%
Apr-25	1.54%	0.72%	5.43%	5.42%	3.42%	2.71%	2.38%	0.16%	0.87%	3.28%
May-25	1.58%	1.02%	5.18%	5.30%	3.34%	3.10%	2.71%	0.57%	0.97%	3.55%
Jun-25	1.29%	1.28%	4.27%	4.40%	3.29%	2.96%	2.76%	0.57%	1.01%	2.43%
Jul-25	1.35%	0.80%	4.56%	4.63%	2.97%	2.24%	2.85%	0.71%	1.07%	2.56%
Aug-25	1.48%	1.07%	3.98%	4.81%	3.10%	2.56%	2.97%	0.63%	1.10%	2.15%
Sep-25	1.71%	1.39%	4.24%	5.10%	3.27%	2.75%	3.20%	0.90%	1.13%	2.56%
Oct-25	1.88%	1.00%	4.70%	5.42%	3.52%	2.26%	3.16%	0.85%	1.20%	3.01%
Nov-25	1.98%	1.32%	5.14%	4.89%	3.68%	2.50%	3.05%	1.04%	1.20%	2.22%
Dec-25	1.64%	1.75%	4.90%	5.18%	3.85%	2.22%	2.98%	1.04%	1.16%	1.93%
Jan-26	1.66%	1.60%	5.30%	5.44%	3.97%	2.52%	3.16%	1.24%	1.18%	2.29%
Feb-26	1.75%	1.40%	5.56%	5.70%	4.04%	2.72%	3.06%	1.44%	1.16%	2.57%
Mar-26	1.97%	1.14%	5.61%	5.66%	4.04%	2.15%	2.71%	1.44%	1.10%	2.67%
Apr-26	2.09%	1.23%	5.74%	5.48%	3.86%	2.18%	2.64%	1.03%	1.07%	2.53%
May-26	2.12%	1.45%	5.90%	5.40%	3.74%	2.39%	2.57%	0.99%	1.13%	2.38%
Jun-26	2.09%	1.44%	6.07%	4.66%	3.62%	2.23%	2.81%	0.86%	1.09%	2.61%

Source: Intex, Morgan Stanley Research

Disclosure Section

Mortgage Backed Securities (MBS) and Collateralized Mortgage Obligations (CMO)

Principal is returned on a monthly basis over the life of the security. Principal prepayment can significantly affect the monthly income stream and the maturity of any type of MBS, including standard MBS, CMOs and Lottery Bonds. Yields and average lives are estimated based on prepayment assumptions and are subject to change based on actual prepayment of the mortgages in the underlying pools. The level of predictability of an MBS/CMO's average life, and its market price, depends on the type of MBS/CMO class purchased and interest rate movements. In general, as interest rates fall, prepayment speeds are likely to increase, thus shortening the MBS/CMO's average life and likely causing its market price to rise. Conversely, as interest rates rise, prepayment speeds are likely to decrease, thus lengthening average life and likely causing the MBS/CMO's market price to fall. Some MBS/CMOs may have "original issue discount" (OID). OID occurs if the MBS/CMO's original issue price is below its stated redemption price at maturity, and results in "imputed interest" that must be reported annually for tax purposes, resulting in a tax liability even though interest was not received. Investors are urged to consult their tax advisors for more information. Government agency backing applies only to the face value of the CMO and not to any premium paid.

The information and opinions in Morgan Stanley Research were prepared by Morgan Stanley & Co. LLC, and/or Morgan Stanley C.T.V.M. S.A., and/or Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., and/or Morgan Stanley Canada Limited. As used in this disclosure section, "Morgan Stanley" includes Morgan Stanley & Co. LLC, Morgan Stanley C.T.V.M. S.A., Morgan Stanley Mexico, Casa de Bolsa, S.A. de C.V., Morgan Stanley Canada Limited and their affiliates as necessary.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Carolyn L. Campbell; James Egan; Raquel Kanner.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of June 30, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.

For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores, Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Carolyn L Campbell; Raquel Kanner; James Egan.

© 2026 Morgan Stanley